Module 6

Modeling of Synchronous Machines

(Chap. 5 of Krause's Book)

Most Important Topics and Concepts

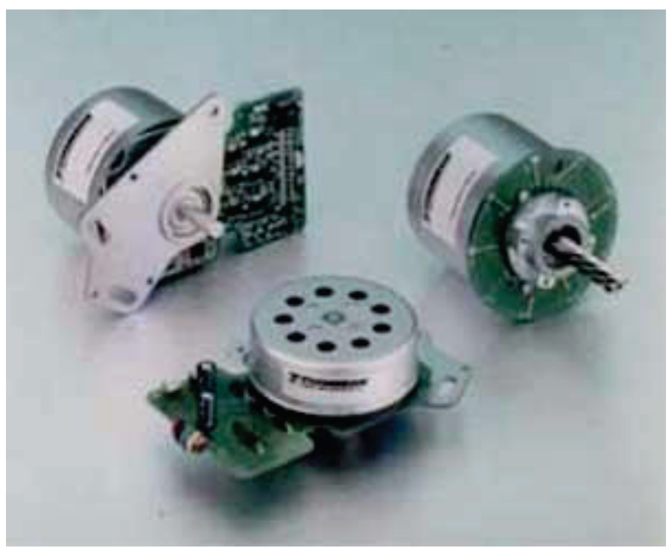
- Types and construction of commonly used synchronous machines
- Steady-state equivalent circuit
- Power conversion, torque-speed characteristics
- Dynamic modeling of ~~A~~SMs for transient analysis

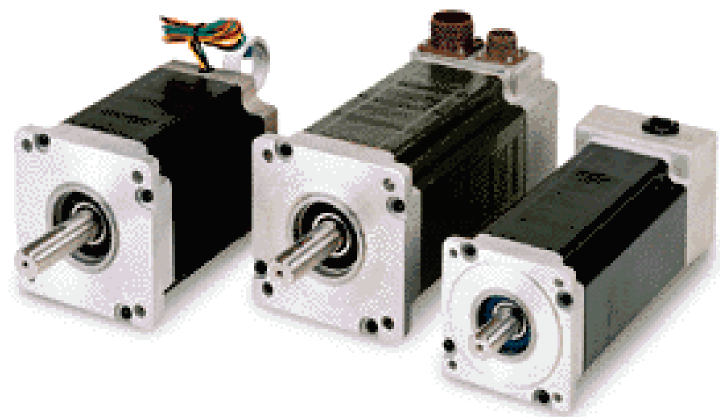
Synchronous Machines

- Portable Diesel Generators 0.5 . . . 10 kW

- Car Alternators

Synchronous

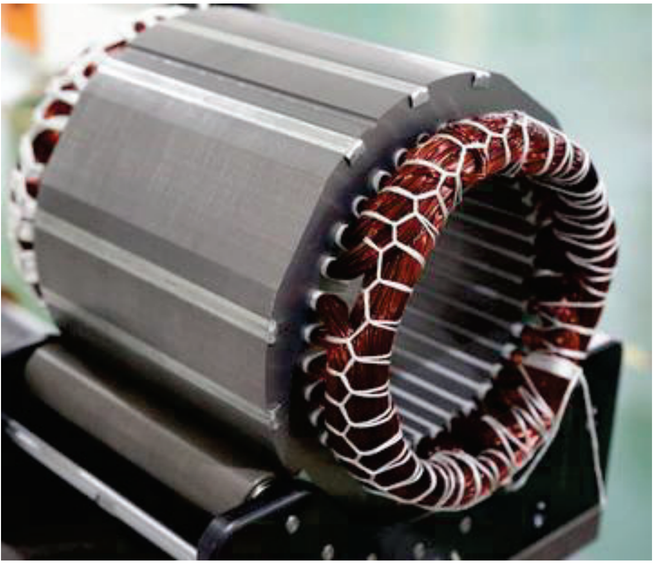


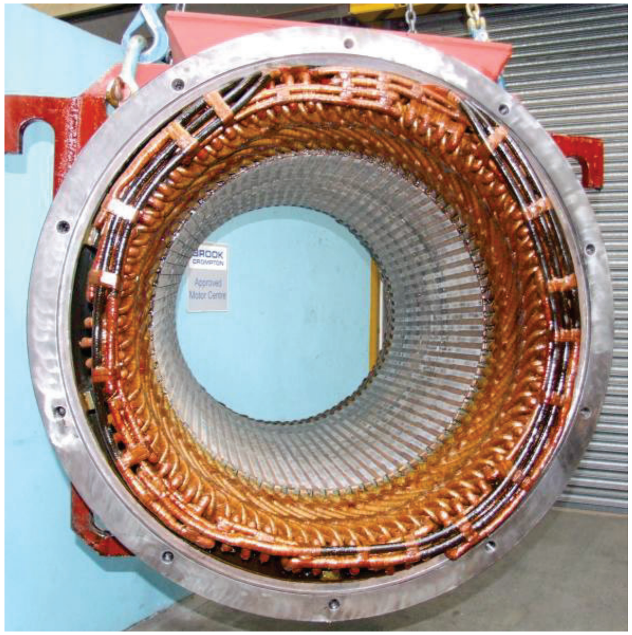


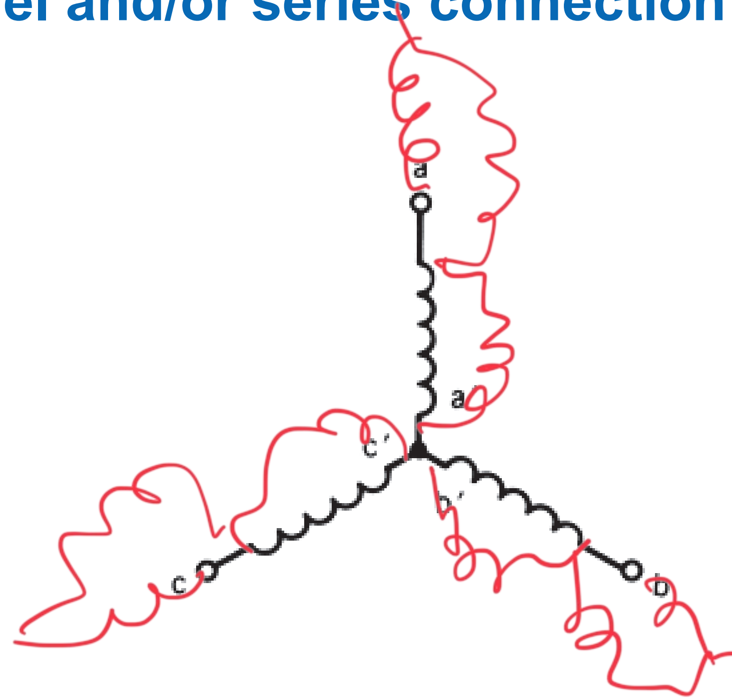
Brushless DC and Servo Motors

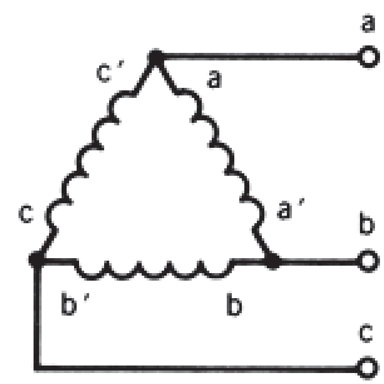
Stator Windings

- ✓ **A typical stator would have 3-phases, which may be Y or Δ connected**
- ✓ **Windings may have multiple sections with parallel and/or series connection**





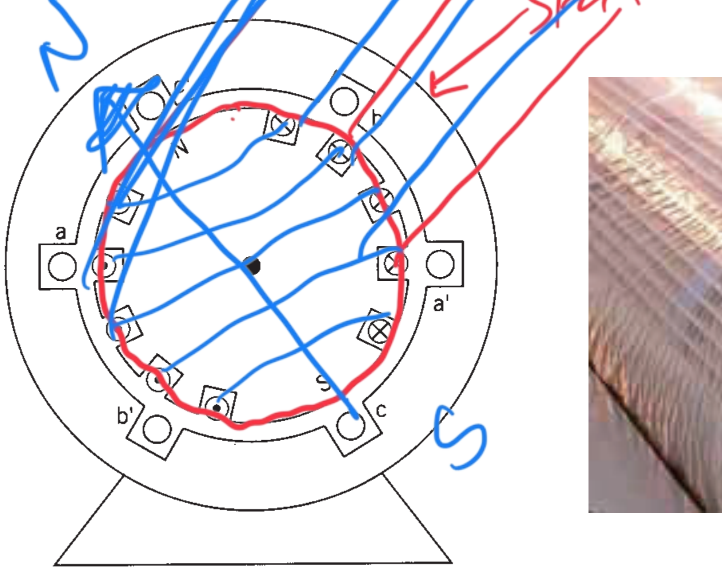




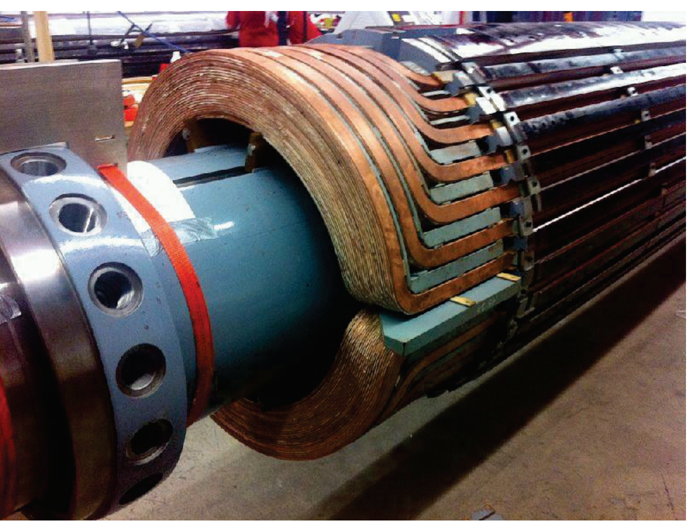
Round Rotor Types

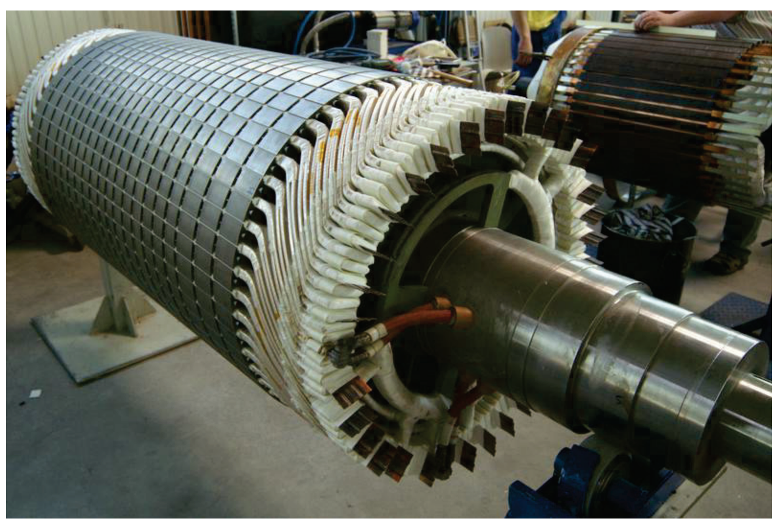
- ✓ **A typical round rotor has small number of magnetic poles (high speed)**

2-Pole Round Rotor



4-Pole Round Rotor



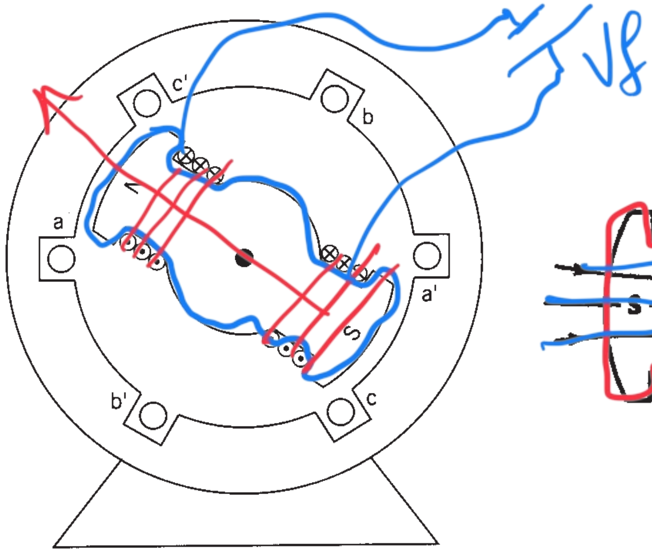


Salient Rotor Types

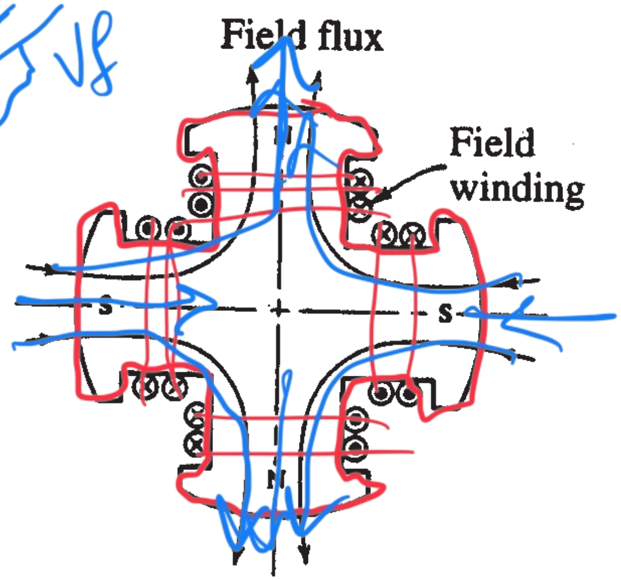
$n_{sy} = \frac{120f}{P}$

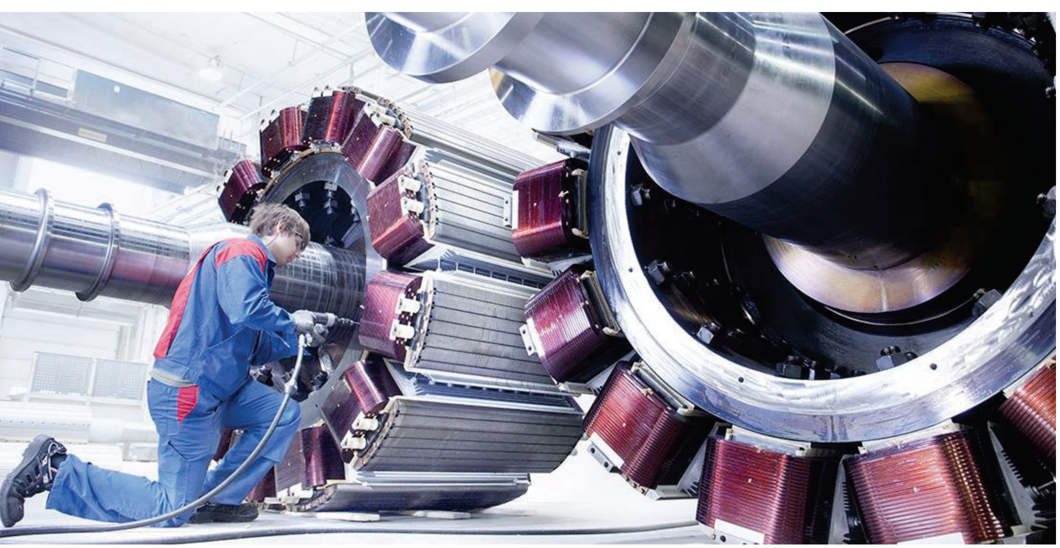
✓ **A typical salient rotor may have more magnetic poles (lower speed)**

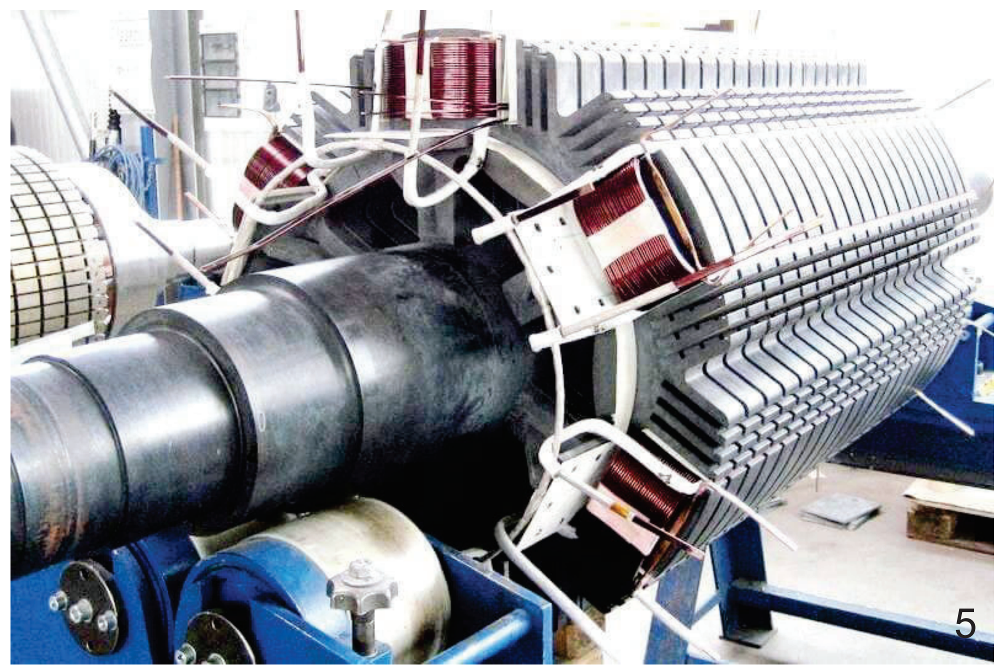
2-Pole Salient Rotor



4-Pole Salient Rotor



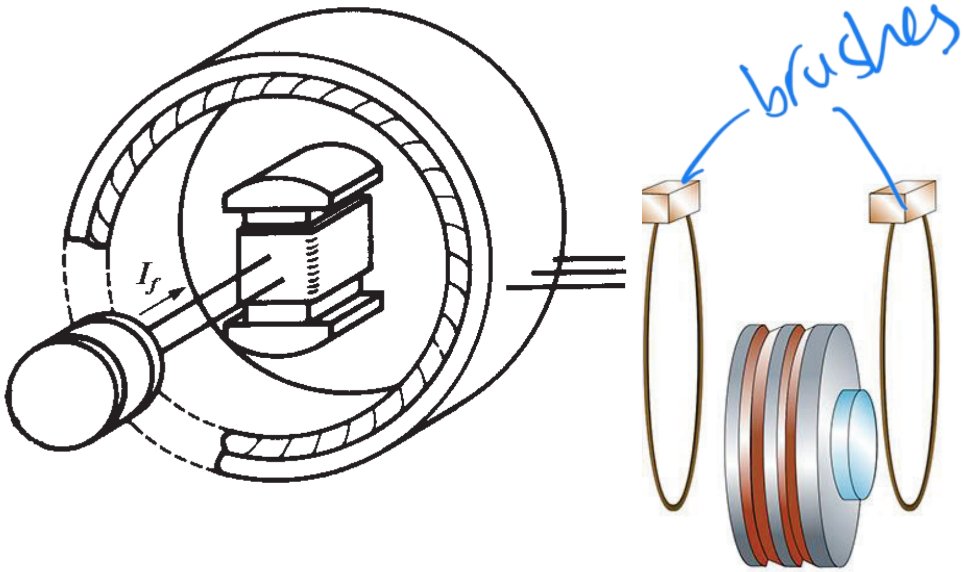


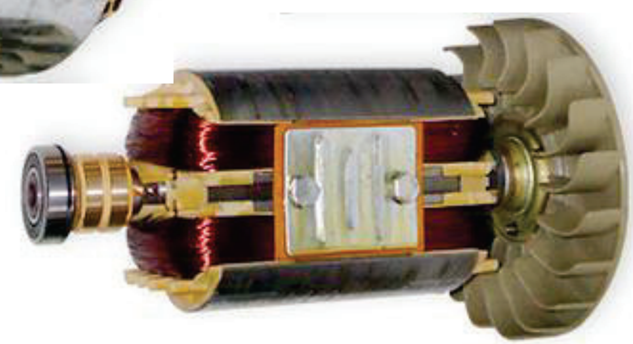


Excitation – Establishing Rotor Field

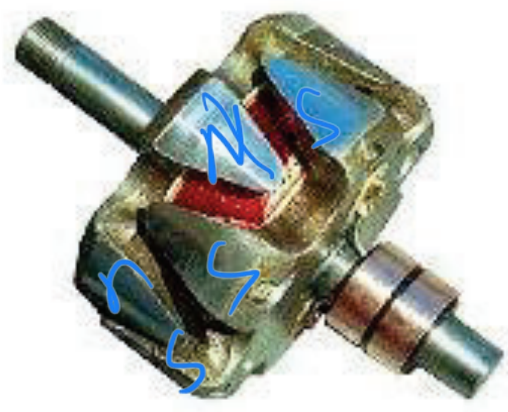
- ✓ **In order to establish the rotor field we need to pass DC current through the field winding**

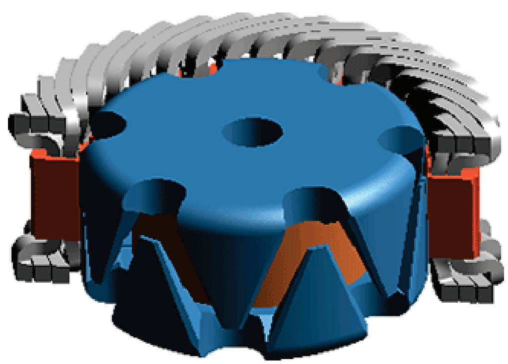
- ✓ Use **Slip-Rings & Brushes** to energize the field winding from external DC source





Claw-Pole Rotor - typically used in automotive alternators

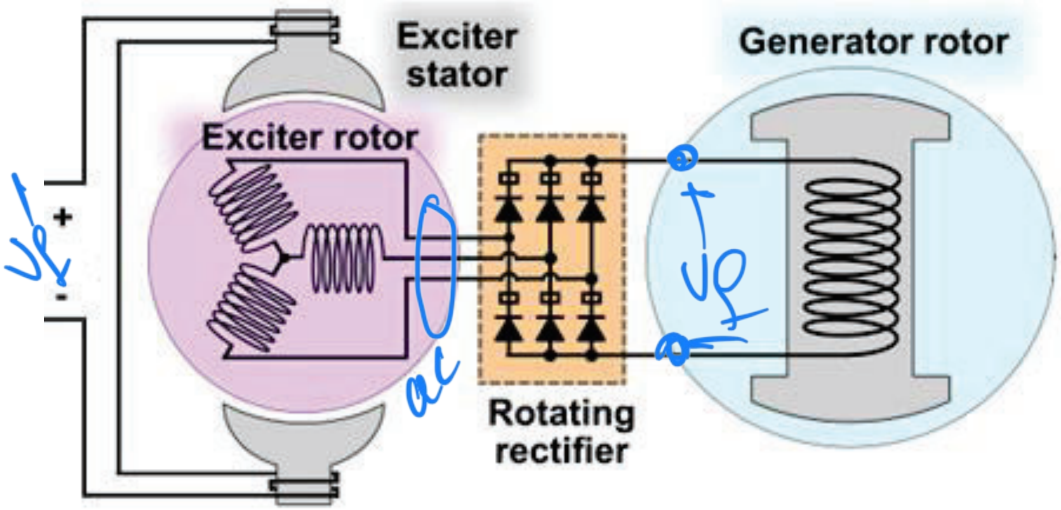


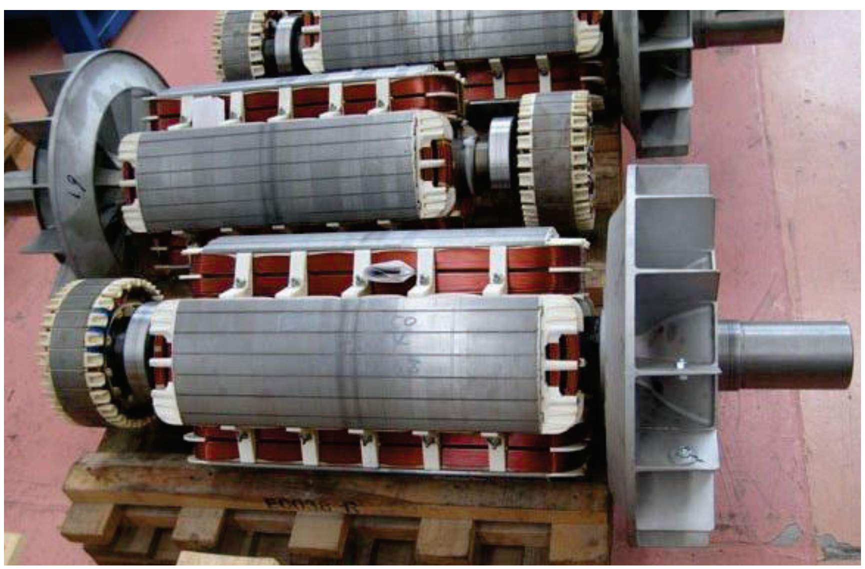


Brushless Excitation – Establishing Field

✓ **In order to establish the rotor field we need to pass DC current through the field winding**

Brushless ➡ Use internal (small) generator and rectifier mounted on the same shaft to produce DC inside the machine

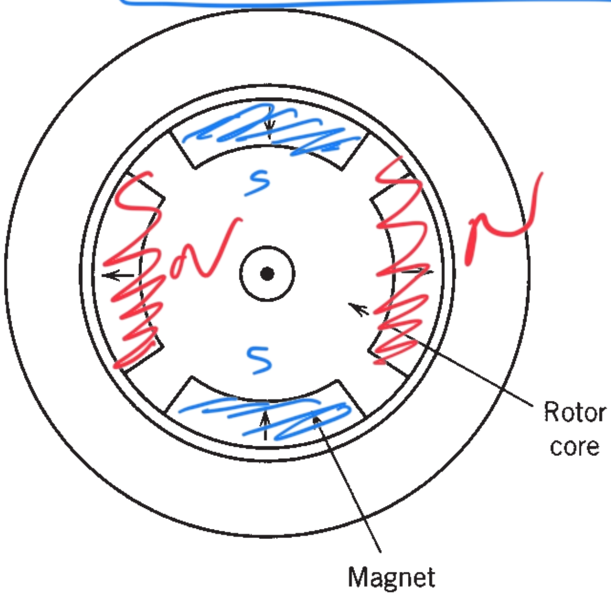


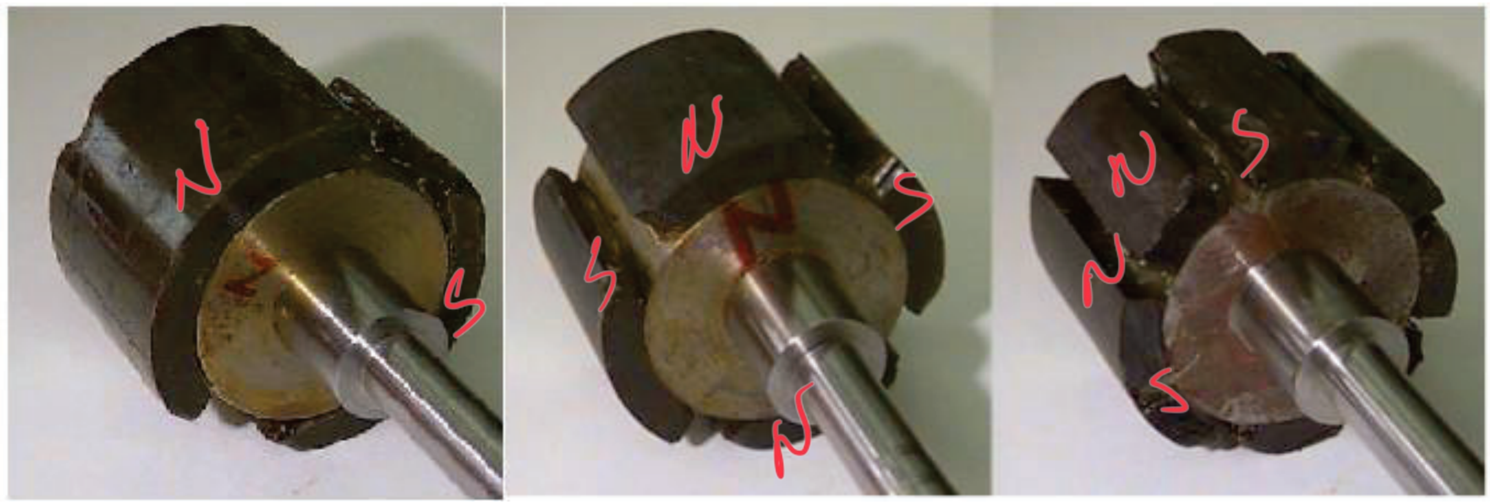


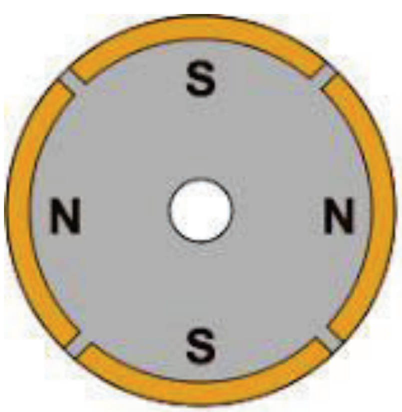
Permanent Magnets (PM) Rotor Types

✓ **Use Permanent Magnets (PMs) to establish the rotor field**

❖ **Surface-Mounted PM (SPM) Rotors**







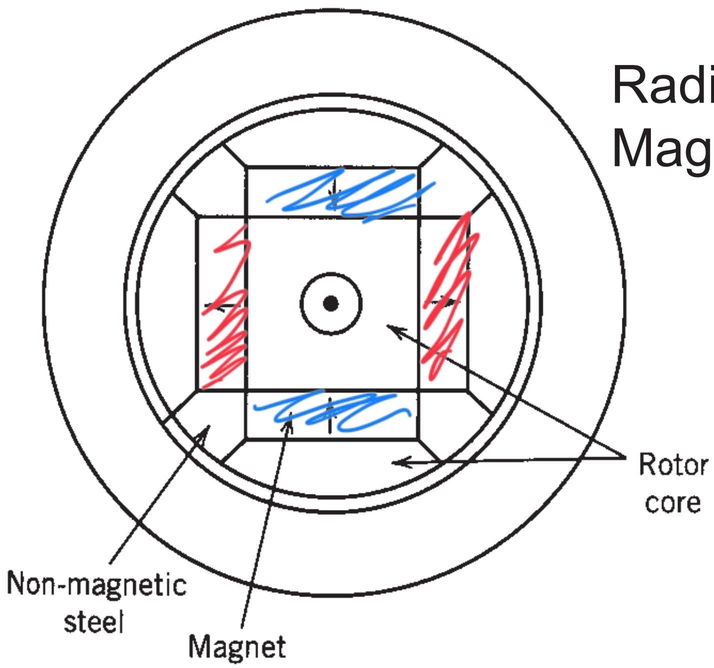
SPM (Surface Permanent Magnet)

Permanent Magnets (PM) Rotor Types

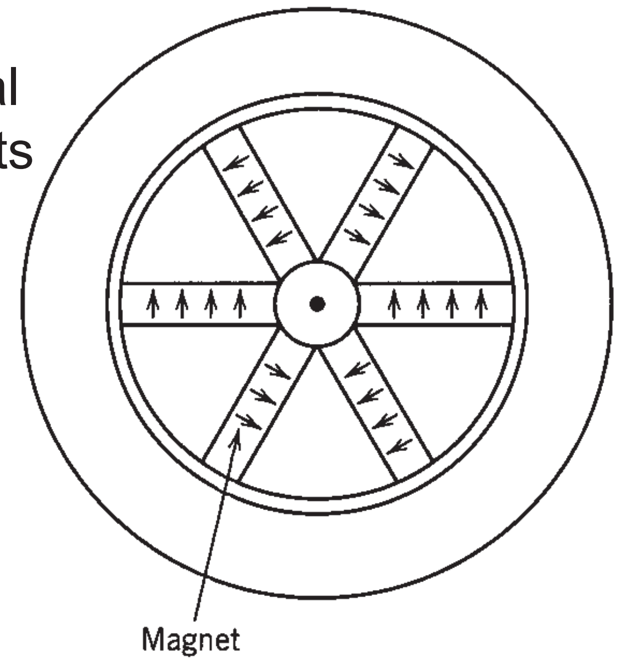
✓ **Use Permanent Magnets (PMs) to establish the rotor field**

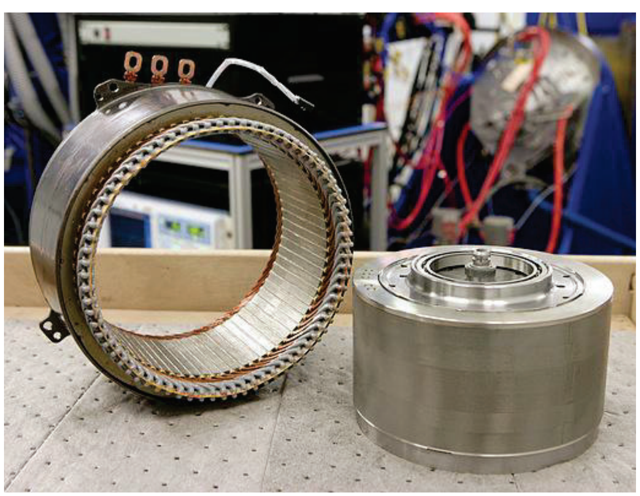
❖ **Interior PM (IPM) Rotors**

Radial Buried Magnets



Circumferential Buried Magnets





IPM (Interior Permanent Magnet)

Reluctance Rotor Types

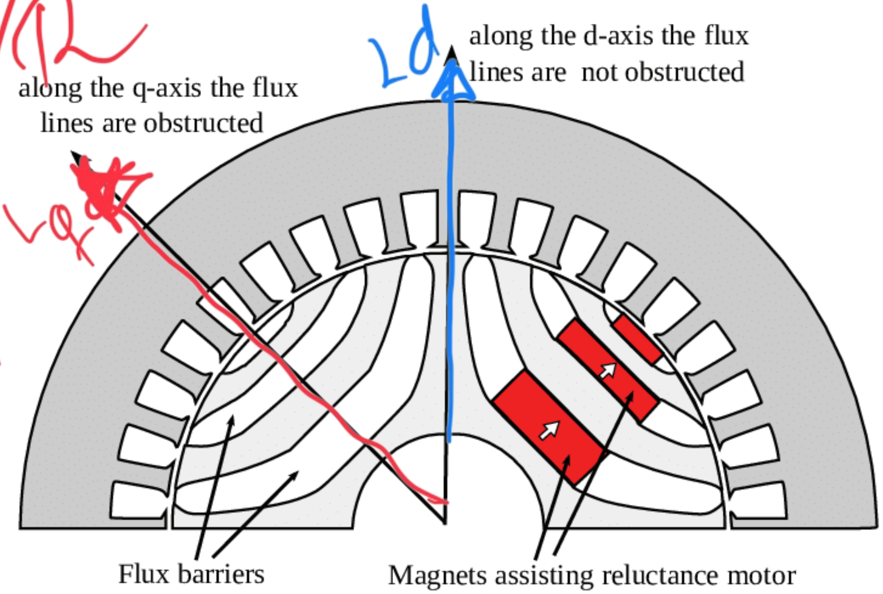
- ✓ Variable reluctance synchronous machines (RSM) use different materials and structures to establish rotor path with small and large reluctance

L_d L_q

$(L_d - L_q)$

$L = N^2/\mathcal{R}$

$L_q \ll L_d$



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Rotating Magnetic Field

1. Given a balanced set of ac currents shifted in time
2. Apply these currents to windings symmetrically shifted in space

⇨ Produce vector $\mathbf{F}_s$ that

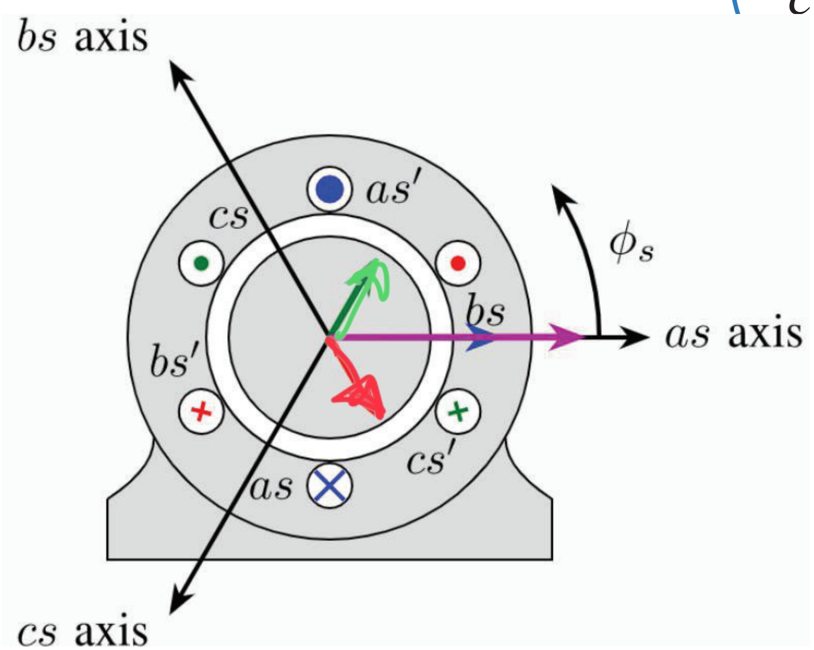
- Has constant magnitude
- Rotates in space

How many phases can you have ?
2, 3, 5, 6, 9, 12, 15 …

$$\begin{cases} i_{as} = I_m \cos(\omega_e t) \\ i_{bs} = I_m \cos(\omega_e t - 90^\circ) \end{cases}$$

$$\begin{cases} i_{as} = I_m \cos(\omega_e t) \\ i_{bs} = I_m \cos(\omega_e t - 120^\circ) \\ i_{cs} = I_m \cos(\omega_e t + 120^\circ) \end{cases}$$

How do you change direction of rotation ?
Reverse the sequence of phases



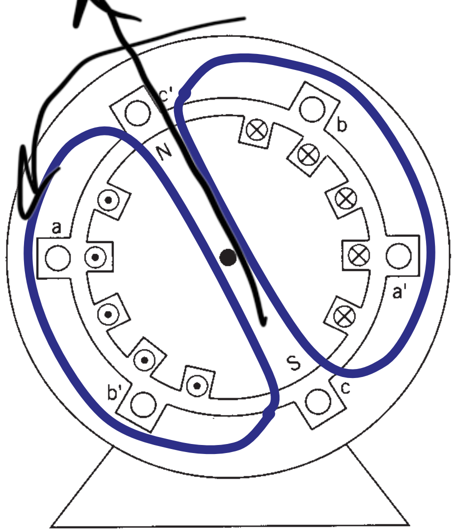
Resulting MMF

$$\mathbf{F}_s = \mathbf{F}_{as} + \mathbf{F}_{bs} + \mathbf{F}_{cs} \quad (\omega t)$$

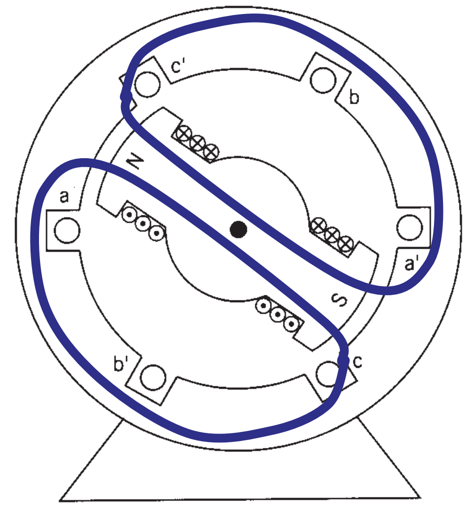
$$\begin{cases} F_{as} = F_m \angle 0^\circ \\ F_{bs} = F_m/2 \angle 60^\circ \\ F_{cs} = F_m/2 \angle -60^\circ \end{cases} = \frac{3}{2} F_m \angle \theta_s$$

Poles & Speed in Syn. Machines

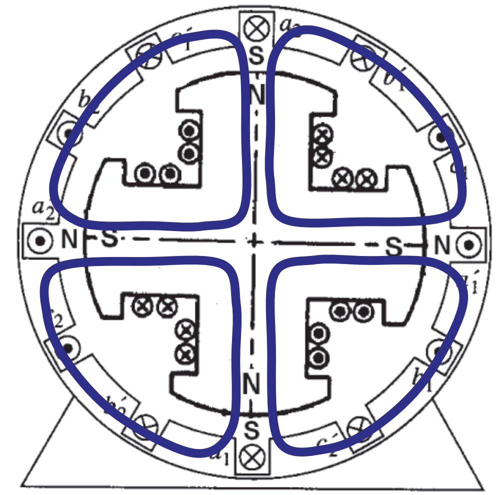
2-Pole Round Rotor



2-Pole Salient Rotor



4-Pole Salient Rotor



In North America

f_e = 60 Hz

$$\omega_e = 2\pi \cdot f_e \text{ [rad/s]}$$

How many magnetic poles can we have?
2, 4, 64…

P pole	n_{syn}	ω_{syn}
2	3600	120π
4	1800	60π
6	1200	40π
8	900	30π
64	112.5	3.75π

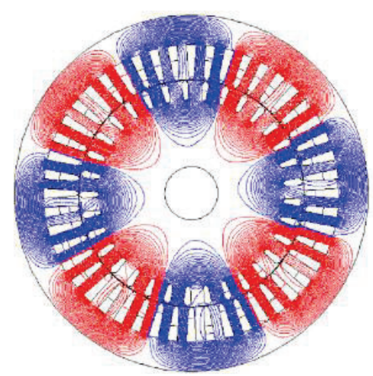
Synchronous Speed
in rev. per min.

$$n_{syn} = \frac{30}{\pi}\omega_{syn} = \frac{120 f_e}{P} \text{ [rpm]}$$

Rotor Poles = # Stator Poles = *P*

Speed $n_{syn} \propto \frac{2}{P}$

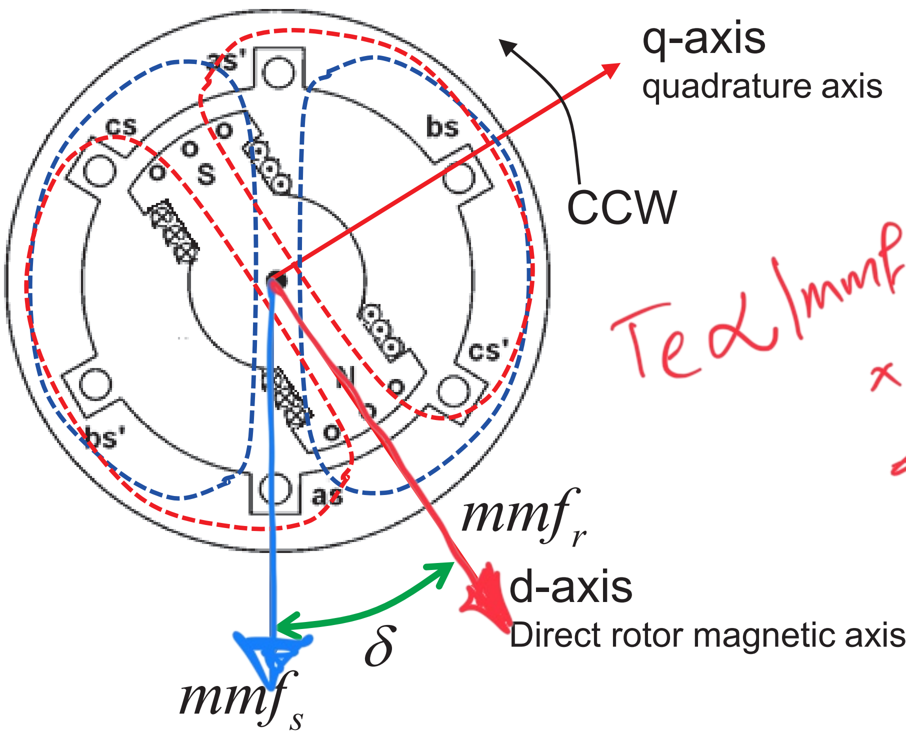
Torque $T_e \propto \frac{P}{2}$



Principle of Operation

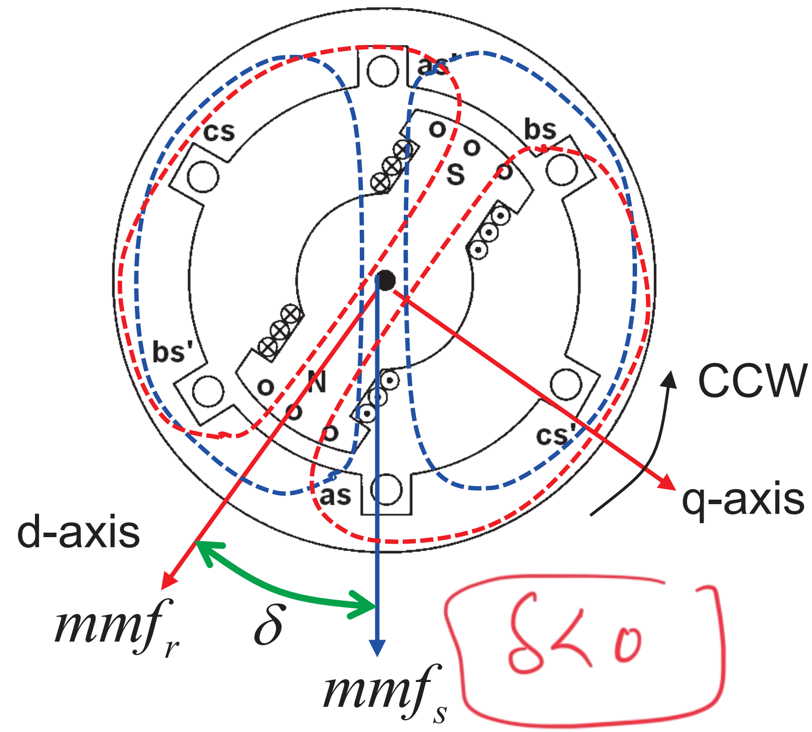
✓ **In steady-state, rotor poles are synchronized with the stator rotating field**

Rotor leads stator rotating field
➡ **Generating Mode**



$T_e \propto |mmf_s||mmf_r| \times \sin\delta$

Rotor lags stator rotating field
➡ **Motoring Mode**



$\boxed{\delta < 0}$

Electrical displacement angle of stator and rotor *MMFs*

$$\begin{cases} \theta_e = \int \omega_e dt \\ \theta_r = \int \omega_r dt \end{cases}$$

Rotor Angle

$$\delta = \theta_r - \theta_e$$

Rotor aligns stator *MMF*
➡ **Idling Mode**

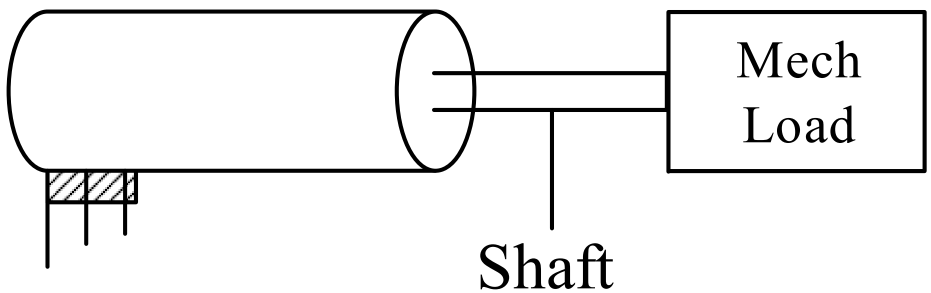
$$T_m = T_e = 0$$

Modeling of Mechanical System

- Recall:

$$\omega_{rm} = \frac{2}{P}\omega_r \qquad \theta_{rm} = \frac{2}{P}\theta_r$$

For *P*-pole Machines



- For simplicity, assume a single rigid body with combined moment of inertia:

$$T_e = J\frac{d\omega_{rm}}{dt} + T_m + T_{fric}$$

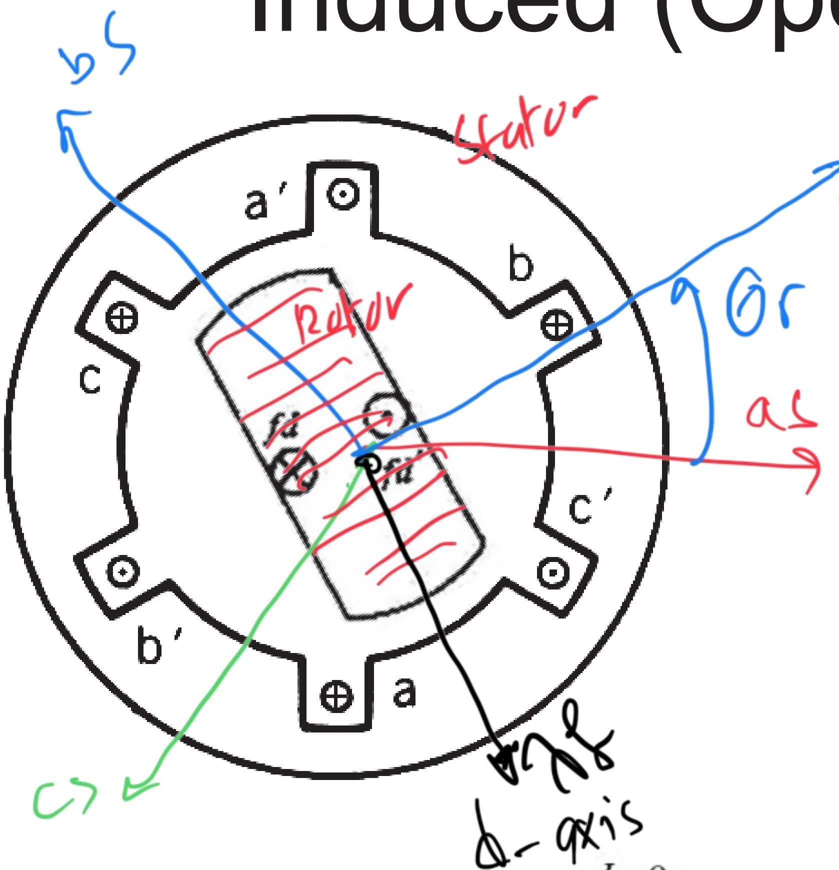
$$J = J_{machine} + J_{load} \quad [kg \cdot m^2]$$

$$\begin{cases} \theta_e = \theta_e(0) + \int_0^t \omega_e \, dt \\ \theta_r = \theta_r(0) + \int_0^t \omega_r \, dt \\ \delta = \theta_r - \theta_e = \int_0^t (\omega_r - \omega_e)\, dt \end{cases}$$

Rotor dynamics:

$$\begin{cases} \left(\frac{P}{2}\right)\frac{d\omega_{rm}}{dt} = \frac{d\omega_r}{dt} = \frac{d^2\delta}{dt^2} \\ J\left(\frac{2}{P}\right)\frac{d^2\delta}{dt^2} = T_e - T_m - T_{fric} \end{cases}$$

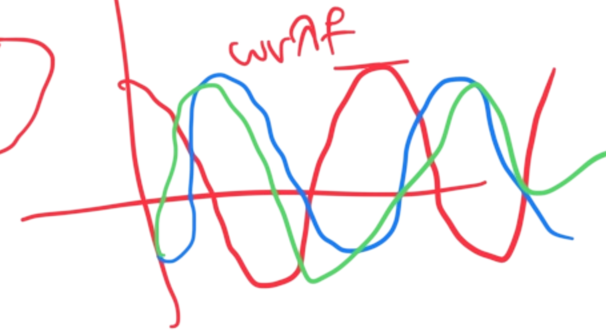
Induced (Open-Circuit) Voltage



$e_a = r_a \cancel{i_a} + d\lambda_a/dt$

$$\boxed{e_a = \frac{d\lambda_a}{dt} = N\frac{d\Phi_a}{dt}}$$

$$\boxed{\lambda_a = \lambda_f \sin(\theta_r)}$$

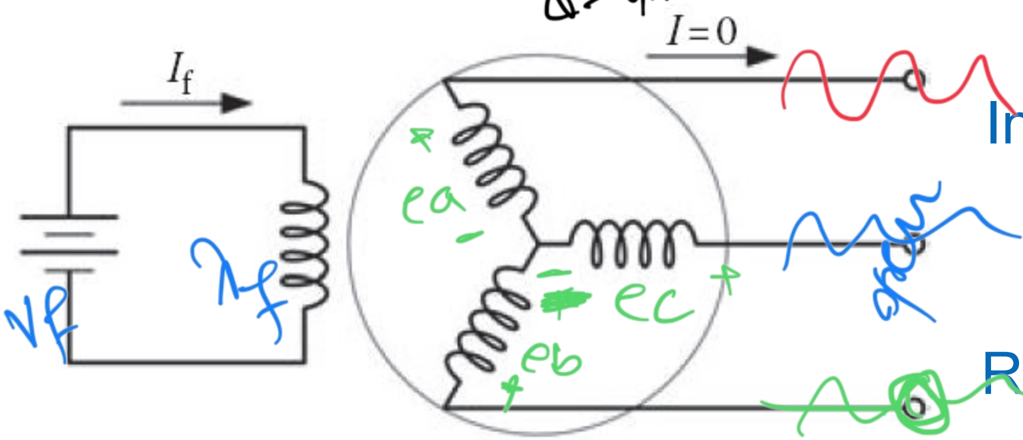


Let $\quad \theta_r = \omega_r t$

$$e_a = \frac{d\lambda_a}{dt} = \omega_r \lambda_f \cos(\theta_r)$$

In Steady-State $\quad \omega_r = \omega_e = 2\pi f_e$

RMS value $\quad E_a = \dfrac{\omega_r \lambda_f}{\sqrt{2}} = \sqrt{2}\pi f_e \lambda_f$



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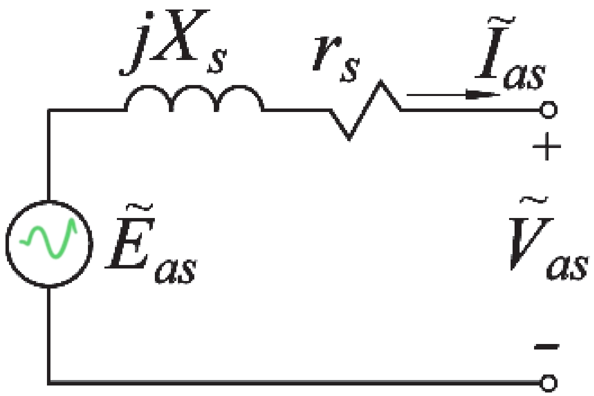
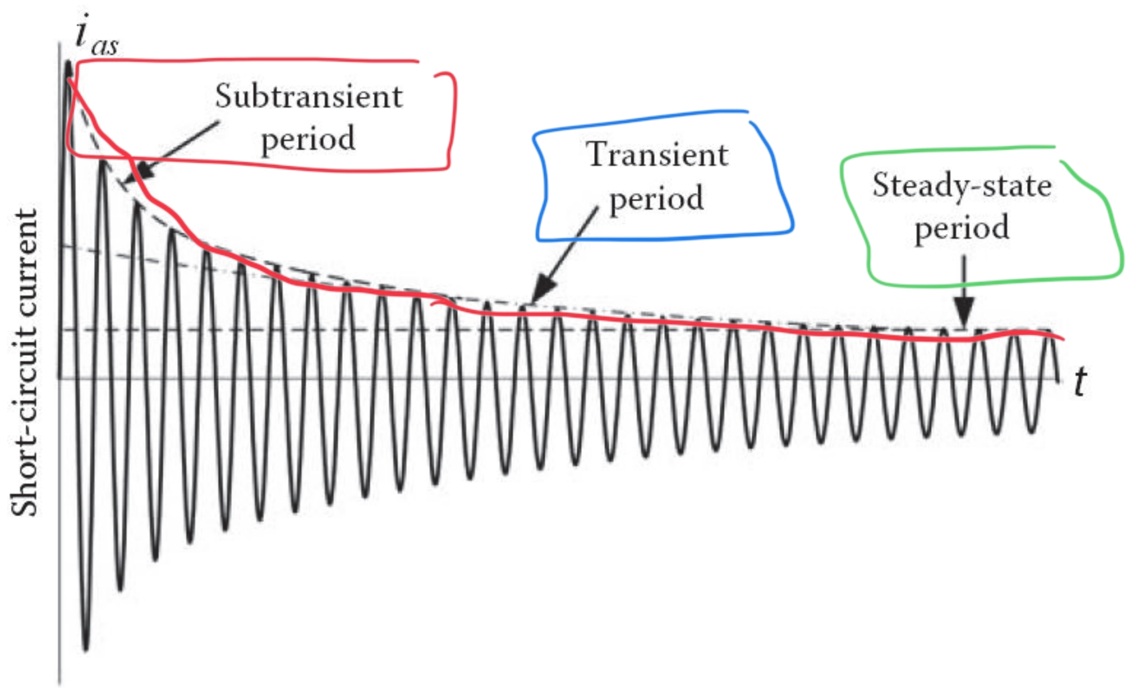
Steady-State Model

✓ **Per-Phase Equivalent Circuit:**

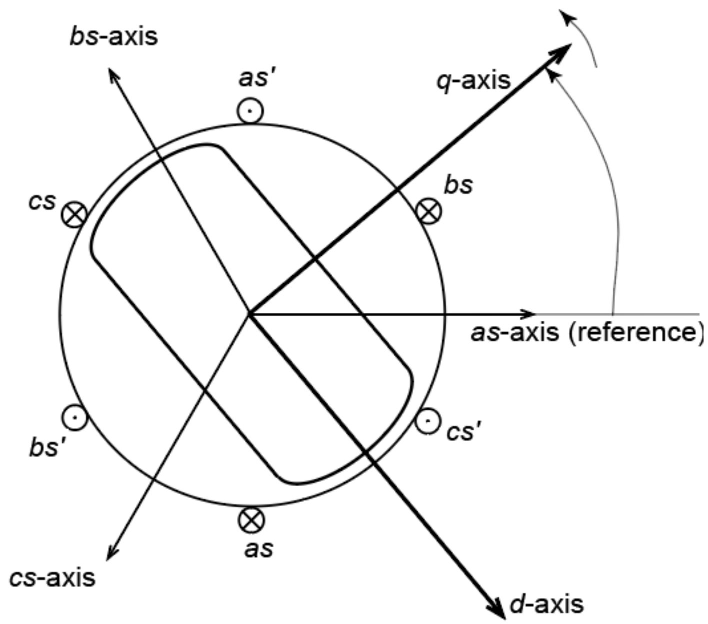


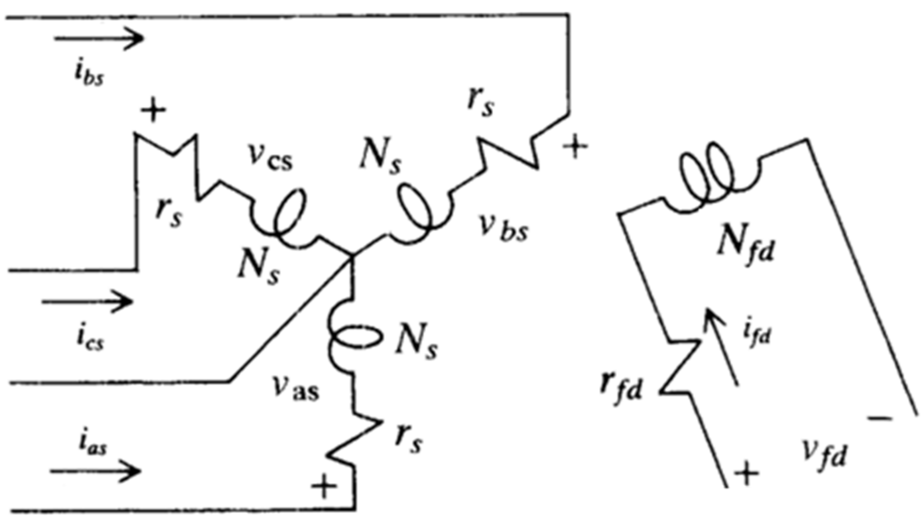
✓ **We need a dynamic model that is good both in steady state to and during transients!**



Modeling of Electric System

- Assume a Salient 2-Pole 3-Phase Symmetric Machine:
  - Assume <u>one</u> field winding:





- **<u>Windings Voltage Equations:</u>**

$$\begin{cases} \mathbf{v}_{abcs} = \mathbf{R}_s \mathbf{i}_{abcs} + \dfrac{d\boldsymbol{\lambda}_{abcs}}{dt} \\ v_{fd} = R_{fd} i_{fd} + \dfrac{d\lambda_{fd}}{dt} \end{cases}$$

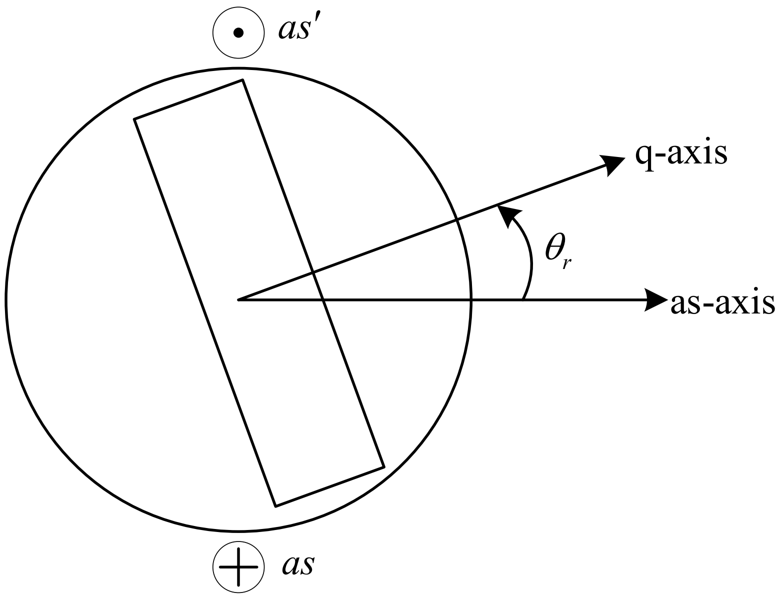
- **<u>Flux Linkage Equations:</u>**

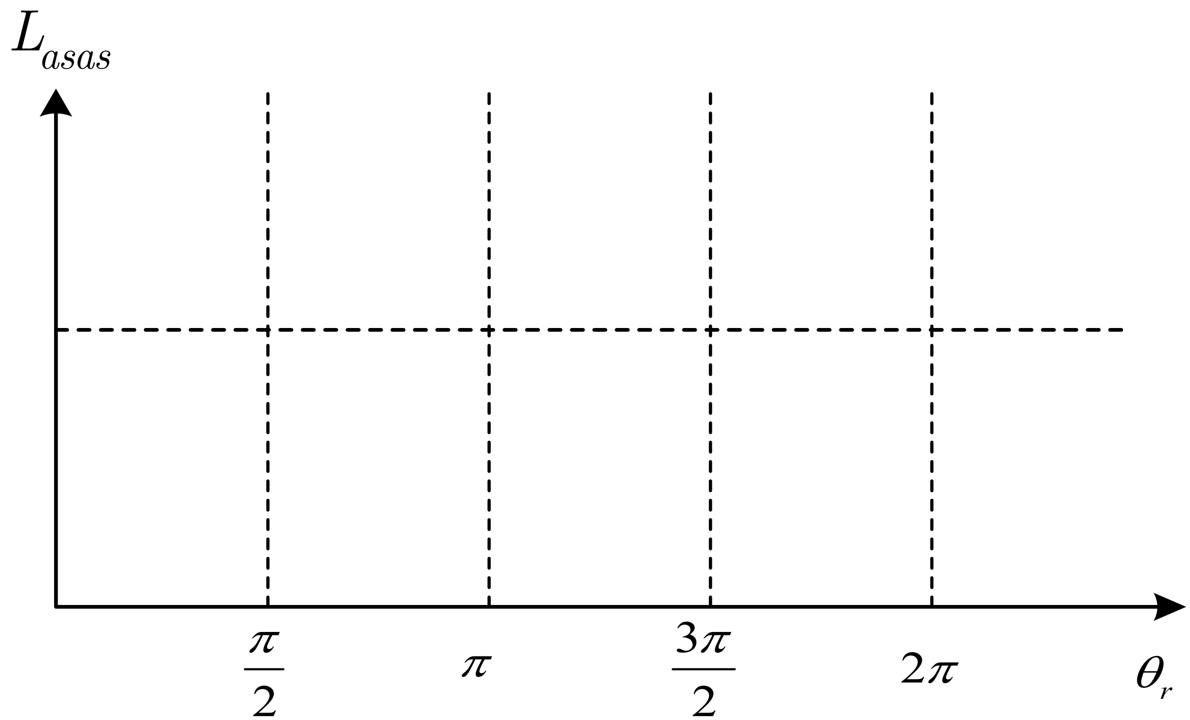
$$\begin{cases} \boldsymbol{\lambda}_{abcs} = \mathbf{L}_s \mathbf{i}_{abcs} + \mathbf{L}_{sr} i_{fd} \\ \lambda_{fd} = \left(\mathbf{L}_{sr}\right)^T \mathbf{i}_{abcs} + L_{fdfd} i_{fd} \end{cases}$$

Modeling of Electric System

❖ System Inductances:

❑ Stator Self Inductances:





$$L_{asas} = L_{ls} +$$

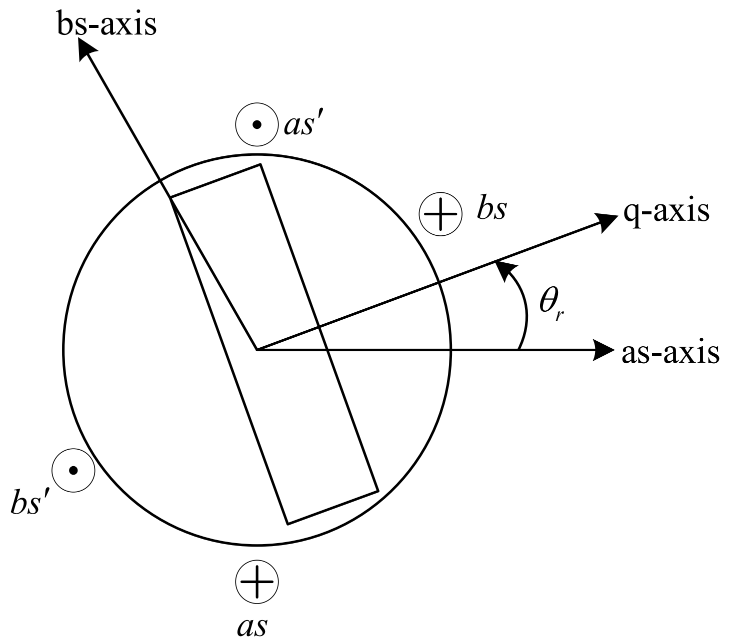
$$L_{bsbs} = L_{ls} +$$

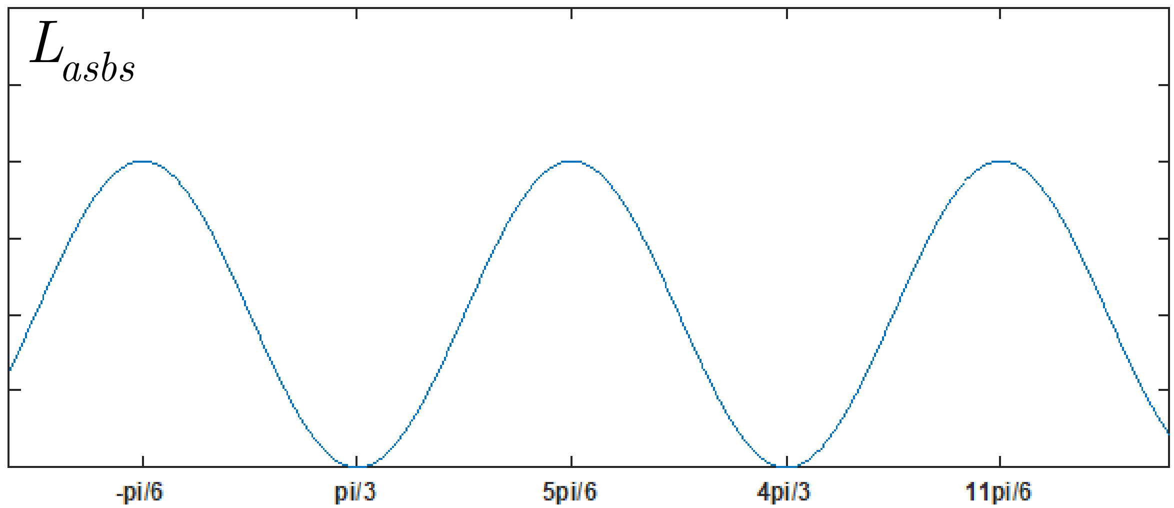
$$L_{cscs} = L_{ls} +$$

Modeling of Electric System

❖ System Inductances:

❑ Stator Mutual Inductances:





$$L_{asbs} =$$

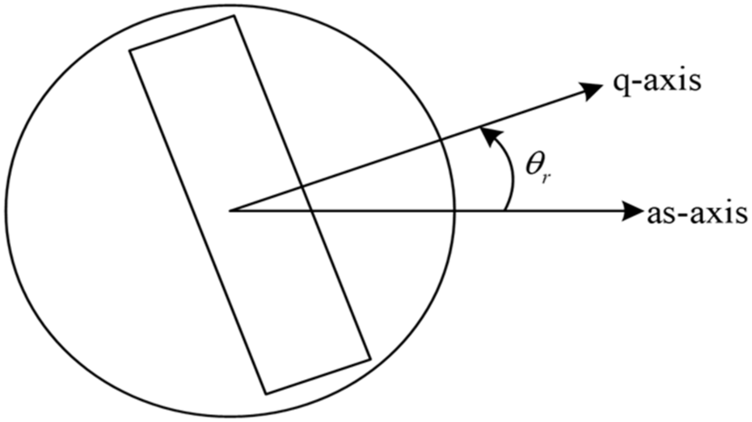
$$L_{ascs} =$$

$$L_{bscs} =$$

Modeling of Electric System

❖ System Inductances:

❑ Rotor Self Inductance:



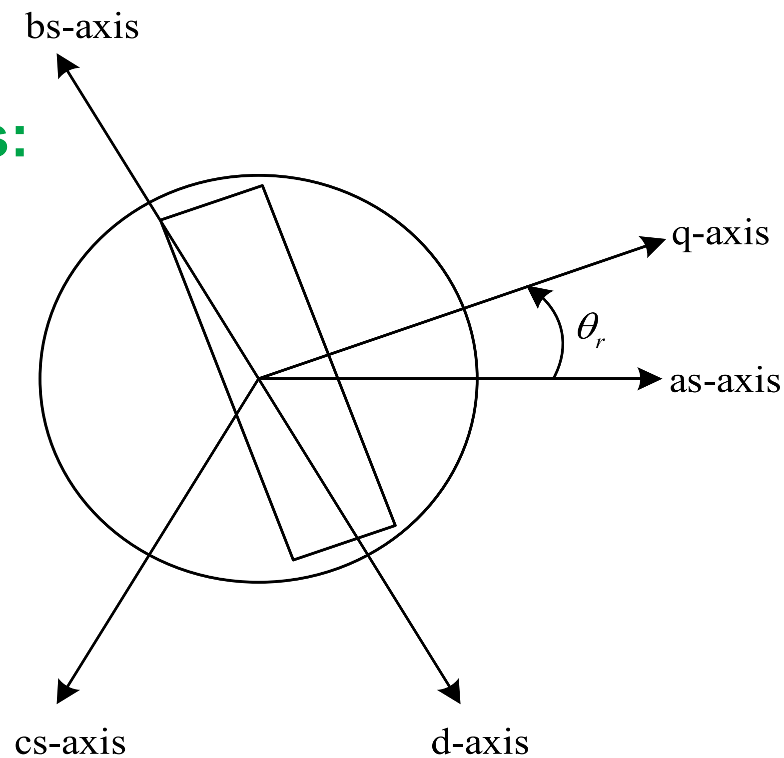
L_{lfd} leakage

L_{mfd} magnetizing

$$L_{fdfd} = L_{lfd} + L_{mfd}$$

❑ Stator-to-Rotor Mutual Inductances:

$$\begin{cases} L_{asfd} = L_{sfd} \cdot \sin\theta_r \\ L_{bsfd} = L_{sfd} \cdot \sin\left(\theta_r - \dfrac{2\pi}{3}\right) \\ L_{csfd} = L_{sfd} \cdot \sin\left(\theta_r + \dfrac{2\pi}{3}\right) \end{cases}$$



Modeling of Electric System

❑ Assume one field winding:

➢ Windings Voltage Equations:

$$\begin{cases} \mathbf{v}_{abcs} = \mathbf{R}_s \mathbf{i}_{abcs} + \dfrac{d\boldsymbol{\lambda}_{abcs}}{dt} \\ v_{fd} = R_{fd} i_{fd} + \dfrac{d\lambda_{fd}}{dt} \end{cases}$$

➢ Flux Linkage Equations:

$$\begin{cases} \boldsymbol{\lambda}_{abcs} = \mathbf{L}_s \mathbf{i}_{abcs} + \mathbf{L}_{sr} i_{fd} \\ \lambda_{fd} = \left(\mathbf{L}_{sr}\right)^T \mathbf{i}_{abcs} + L_{fdfd} i_{fd} \end{cases}$$

$$\mathbf{L}_s = \left[\begin{array}{c:c:c} L_{ls} + L_A - L_B \cos 2\left(\theta_r\right) & -\frac{1}{2}L_A - L_B \cos 2\left(\theta_r - \frac{\pi}{3}\right) & -\frac{1}{2}L_A - L_B \cos 2\left(\theta_r + \frac{\pi}{3}\right) \\ -\frac{1}{2}L_A - L_B \cos 2\left(\theta_r - \frac{\pi}{3}\right) & L_{ls} + L_A - L_B \cos 2\left(\theta_r - \frac{2\pi}{3}\right) & -\frac{1}{2}L_A - L_B \cos 2\left(\theta_r + \pi\right) \\ -\frac{1}{2}L_A - L_B \cos 2\left(\theta_r + \frac{\pi}{3}\right) & -\frac{1}{2}L_A - L_B \cos 2\left(\theta_r + \pi\right) & L_{ls} + L_A - L_B \cos 2\left(\theta_r + \frac{2\pi}{3}\right) \end{array}\right]$$

$$\mathbf{L}_{sr}^T = L_{sfd}\left[\sin\left(\theta_r\right) \quad \sin\left(\theta_r - \frac{2\pi}{3}\right) \quad \sin\left(\theta_r + \frac{2\pi}{3}\right)\right]$$

Modeling of Electric System

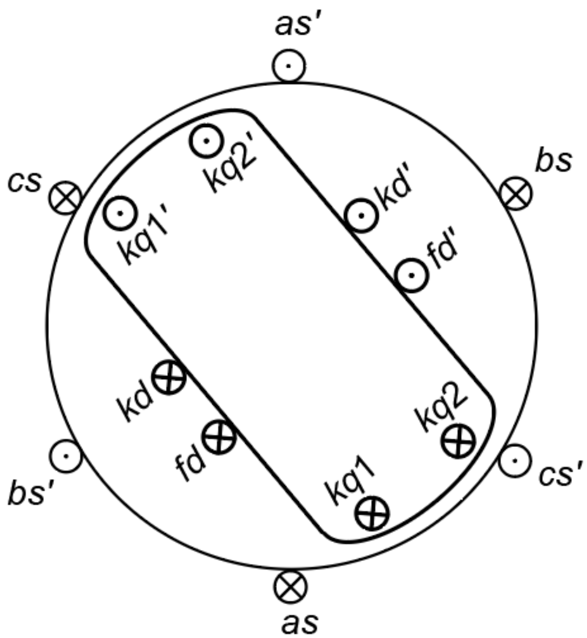
❑ Full Order Model:

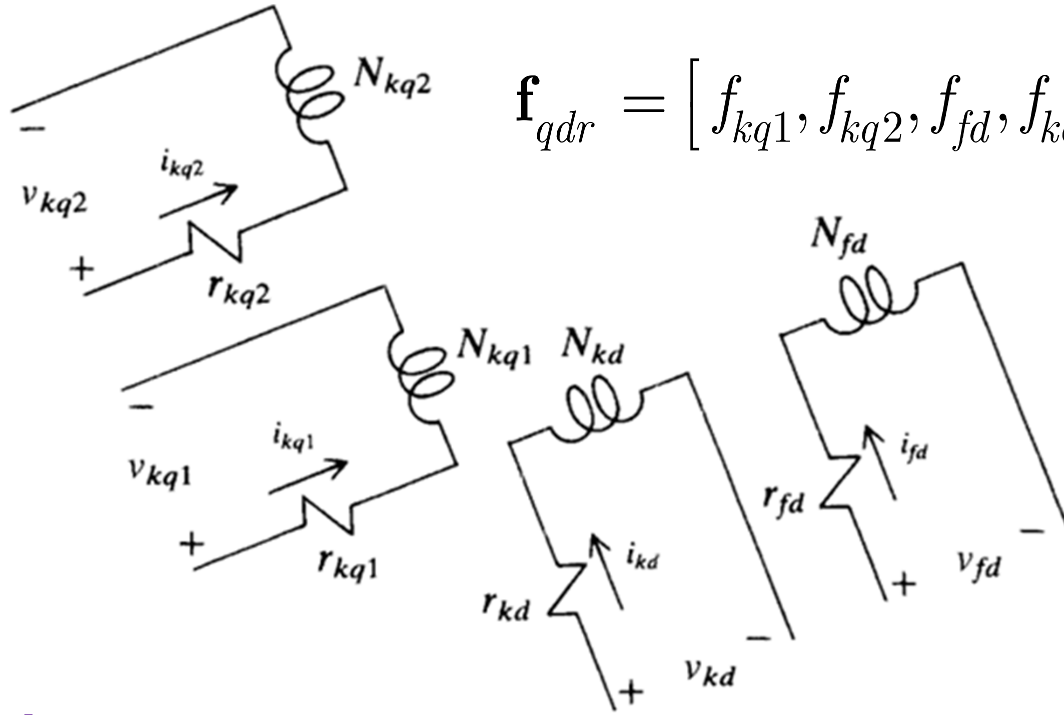
❖ Assume 3 damper windings on rotor:

✓ *fd* - field winding on *d*-axis

✓ *kd* - damper winding on *d*-axis

✓ *kq*1 and *kq*2 - damper windings on *q*-axis





$$\mathbf{f}_{qdr} = \left[f_{kq1}, f_{kq2}, f_{fd}, f_{kd}\right]^T$$

➢ **Windings Voltage Equations:**

$$\begin{cases} \mathbf{v}_{abcs} = \mathbf{R}_s \mathbf{i}_{abcs} + \dfrac{d\boldsymbol{\lambda}_{abcs}}{dt} \\ \mathbf{v}_{qdr} = \mathbf{R}_r \mathbf{i}_{qdr} + \dfrac{d\boldsymbol{\lambda}_{qdr}}{dt} \end{cases}$$

➢ **Flux Linkage Equations:**

$$\begin{cases} \boldsymbol{\lambda}_{abcs} = \mathbf{L}_s \mathbf{i}_{abcs} + \mathbf{L}_{sr} \mathbf{i}_{qdr} \\ \boldsymbol{\lambda}_{qdr} = \mathbf{L}_r \mathbf{i}_{qdr} + \left(\mathbf{L}_{sr}\right)^T \mathbf{i}_{abcs} \end{cases}$$

Modeling of Electric System

❑ Full Order Model:

❖ System Matrices:

➢ **Stator and Rotor Resistance Matrices:**

$$\mathbf{R}_s = \begin{bmatrix} r_s & & \\ & r_s & \\ & & r_s \end{bmatrix} \quad \mathbf{R}_r = \begin{bmatrix} r_{kq1} & & & \\ & r_{kq2} & & \\ & & r_{fd} & \\ & & & r_{kd} \end{bmatrix}$$

➢ **Stator Inductance Matrix:**

$$\mathbf{L}_s = \left[\begin{array}{c:c:c} L_{ls} + L_A - L_B\cos 2(\theta_r) & -\frac{1}{2}L_A - L_B\cos 2\left(\theta_r - \frac{\pi}{3}\right) & -\frac{1}{2}L_A - L_B\cos 2\left(\theta_r + \frac{\pi}{3}\right) \\ -\frac{1}{2}L_A - L_B\cos 2\left(\theta_r - \frac{\pi}{3}\right) & L_{ls} + L_A - L_B\cos 2\left(\theta_r - \frac{2\pi}{3}\right) & -\frac{1}{2}L_A - L_B\cos 2(\theta_r + \pi) \\ -\frac{1}{2}L_A - L_B\cos 2\left(\theta_r + \frac{\pi}{3}\right) & -\frac{1}{2}L_A - L_B\cos 2(\theta_r + \pi) & L_{ls} + L_A - L_B\cos 2\left(\theta_r + \frac{2\pi}{3}\right) \end{array}\right]$$

➢ **Rotor Inductance Matrix:**

$$\mathbf{L}_r = \left|\begin{array}{c:c:c:c} L_{lkq1} + L_{mkq1} & L_{kq1kq2} & & \\ L_{kq2kq1} & L_{lkq2} + L_{mkq2} & & \\ & & L_{lfd} + L_{mfd} & L_{fdkd} \\ & & L_{kdfd} & L_{lkd} + L_{mkd} \end{array}\right|$$

➢ **Stator to Rotor Mutual Inductance Matrix:**

$$\mathbf{L}_{sr} = \left|\begin{array}{c:c:c:c} L_{skq1}\cos(\theta_r) & L_{skq2}\cos(\theta_r) & L_{sfd}\sin(\theta_r) & L_{skd}\sin(\theta_r) \\ L_{skq1}\cos\left(\theta_r - \frac{2\pi}{3}\right) & L_{skq2}\cos\left(\theta_r - \frac{2\pi}{3}\right) & L_{sfd}\sin\left(\theta_r - \frac{2\pi}{3}\right) & L_{skd}\sin\left(\theta_r - \frac{2\pi}{3}\right) \\ L_{skq1}\cos\left(\theta_r + \frac{2\pi}{3}\right) & L_{skq2}\cos\left(\theta_r + \frac{2\pi}{3}\right) & L_{sfd}\sin\left(\theta_r + \frac{2\pi}{3}\right) & L_{skd}\sin\left(\theta_r + \frac{2\pi}{3}\right) \end{array}\right|$$

Modeling of Electric System

❑ Full Order Model:

❖ System Matrices:

$$L_{mkq1} = \left(\frac{N_{kq1}}{N_s}\right)^2 \left(\frac{2}{3}\right) L_{mq} \quad L_{mfd} = \left(\frac{N_{fd}}{N_s}\right)^2 \left(\frac{2}{3}\right) L_{md} \quad L_{kq1kq2} = L_{kq2kq1} = \left(\frac{N_{kq1}N_{kq2}}{N_s^2}\right)\left(\frac{2}{3}\right) L_{mq}$$

$$L_{mkq2} = \left(\frac{N_{kq2}}{N_s}\right)^2 \left(\frac{2}{3}\right) L_{mq} \quad L_{mkd} = \left(\frac{N_{kd}}{N_s}\right)^2 \left(\frac{2}{3}\right) L_{md} \quad L_{fdkd} = L_{kdfd} = \left(\frac{N_{fd}N_{kd}}{N_s^2}\right)\left(\frac{2}{3}\right) L_{md}$$

$$L_{skq1} = \left(\frac{N_{kq1}}{N_s}\right)\left(\frac{2}{3}\right) L_{mq} \qquad L_{sfd} = \left(\frac{N_{fd}}{N_s}\right)\left(\frac{2}{3}\right) L_{md}$$

$$L_{skq2} = \left(\frac{N_{kq2}}{N_s}\right)\left(\frac{2}{3}\right) L_{mq} \qquad L_{skd} = \left(\frac{N_{kd}}{N_s}\right)\left(\frac{2}{3}\right) L_{md}$$

magnetizing inductances along *q* and *d* axes

$$L_{mq} = \frac{3}{2}\left(L_A - L_B\right) \qquad L_{md} = \frac{3}{2}\left(L_A + L_B\right)$$

Modeling of Electric System

❖ Coupled-Circuit Phase Domain (CCPD) Model:

➢ **Dynamic Equations:**

$$\frac{d}{dt}\begin{bmatrix}\boldsymbol{\lambda}_{abcs}\\ \boldsymbol{\lambda}_{qdr}\end{bmatrix} = \begin{bmatrix}-\mathbf{R}_s & \\ & -\mathbf{R}_r\end{bmatrix}\begin{bmatrix}\mathbf{i}_{abcs}\\ \mathbf{i}_{qdr}\end{bmatrix} + \begin{bmatrix}\mathbf{v}_{abcs}\\ \mathbf{v}_{qdr}\end{bmatrix} \qquad \begin{bmatrix}\boldsymbol{\lambda}_{abcs}\\ \boldsymbol{\lambda}_{qdr}\end{bmatrix} = \begin{bmatrix}\mathbf{L}_s & \mathbf{L}_{sr}\\ \left(\mathbf{L}_{sr}\right)^T & \mathbf{L}_r\end{bmatrix}\begin{bmatrix}\mathbf{i}_{abcs}\\ \mathbf{i}_{qdr}\end{bmatrix}$$

➢ Refer all rotor variables to the stator side by appropriate turns ratios

$$\begin{cases} v_j' = \left(\dfrac{N_s}{N_j}\right) \times v_j \\[2ex] \lambda_j' = \left(\dfrac{N_s}{N_j}\right) \times \lambda_j \\[2ex] i_j' = \left(\dfrac{2}{3}\right)\left(\dfrac{N_j}{N_s}\right) \times i_j \end{cases}$$

For $j = kq1, kq2, fd, kd$

$$\begin{cases} r_j' = \left(\dfrac{3}{2}\right)\left(\dfrac{N_s}{N_j}\right)^2 \times r_j \\[2ex] L_j' = \left(\dfrac{3}{2}\right)\left(\dfrac{N_s}{N_r}\right)^2 \times L_j \end{cases}$$

Modeling of Electric System

❑ Full Order Model:

❖ System Matrices:

- **Stator and Rotor Resistance Matrices:**

$$\mathbf{R}_s = \begin{bmatrix} r_s & & \\ & r_s & \\ & & r_s \end{bmatrix} \quad \mathbf{R}'_r = \begin{bmatrix} r'_{kq1} & & & \\ & r'_{kq2} & & \\ & & r'_{fd} & \\ & & & r'_{kd} \end{bmatrix}$$

- **Stator Inductance Matrix:**

$$\mathbf{L}_s = \begin{bmatrix} L_{ls} + L_A - L_B\cos 2\left(\theta_r\right) & -\frac{1}{2}L_A - L_B\cos 2\left(\theta_r - \frac{\pi}{3}\right) & -\frac{1}{2}L_A - L_B\cos 2\left(\theta_r + \frac{\pi}{3}\right) \\ -\frac{1}{2}L_A - L_B\cos 2\left(\theta_r - \frac{\pi}{3}\right) & L_{ls} + L_A - L_B\cos 2\left(\theta_r - \frac{2\pi}{3}\right) & -\frac{1}{2}L_A - L_B\cos 2\left(\theta_r + \pi\right) \\ -\frac{1}{2}L_A - L_B\cos 2\left(\theta_r + \frac{\pi}{3}\right) & -\frac{1}{2}L_A - L_B\cos 2\left(\theta_r + \pi\right) & L_{ls} + L_A - L_B\cos 2\left(\theta_r + \frac{2\pi}{3}\right) \end{bmatrix}$$

- **Rotor Inductance Matrix:**

$$\mathbf{L}'_r = \begin{bmatrix} L'_{lkq1} + L_{mq} & L_{mq} & & \\ L_{mq} & L'_{kq2} + L_{mq} & & \\ & & L'_{lfd} + L_{md} & L_{md} \\ & & L_{md} & L_{lkd} + L_{md} \end{bmatrix}$$

- **Stator to Rotor Mutual Inductance Matrix:**

$$\mathbf{L}'_{sr} = \begin{bmatrix} L_{mq}\cos\left(\theta_r\right) & L_{mq}\cos\left(\theta_r\right) & L_{md}\sin\left(\theta_r\right) & L_{md}\sin\left(\theta_r\right) \\ L_{mq}\cos\left(\theta_r - \frac{2\pi}{3}\right) & L_{mq}\cos\left(\theta_r - \frac{2\pi}{3}\right) & L_{md}\sin\left(\theta_r - \frac{2\pi}{3}\right) & L_{md}\sin\left(\theta_r - \frac{2\pi}{3}\right) \\ L_{mq}\cos\left(\theta_r + \frac{2\pi}{3}\right) & L_{mq}\cos\left(\theta_r + \frac{2\pi}{3}\right) & L_{md}\sin\left(\theta_r + \frac{2\pi}{3}\right) & L_{md}\sin\left(\theta_r + \frac{2\pi}{3}\right) \end{bmatrix}$$

Modeling of Electric System

❖ Coupled-Circuit Phase Domain (CCPD) Model:

➢ Dynamic Equations:

$$\frac{d}{dt}\begin{bmatrix}\boldsymbol{\lambda}_{abcs}\\ \boldsymbol{\lambda}'_{qdr}\end{bmatrix}=\begin{bmatrix}-\mathbf{R}_s & \\ & -\mathbf{R}'_r\end{bmatrix}\begin{bmatrix}\mathbf{i}_{abcs}\\ \mathbf{i}'_{qdr}\end{bmatrix}+\begin{bmatrix}\mathbf{v}_{abcs}\\ \mathbf{v}'_{qdr}\end{bmatrix}\qquad \begin{bmatrix}\boldsymbol{\lambda}_{abcs}\\ \boldsymbol{\lambda}'_{abcr}\end{bmatrix}=\begin{bmatrix}\mathbf{L}_s & \mathbf{L}'_{sr}\\ (2/3)\left(\mathbf{L}'_{sr}\right)^T & \mathbf{L}'_r\end{bmatrix}\begin{bmatrix}\mathbf{i}_{abcs}\\ \mathbf{i}'_{qdr}\end{bmatrix}$$

➢ State-Space Form:

➢ Model Properties:

Coupled-Circuit State-Space Model

$$\left\{\begin{array}{l}1)\\ 2)\\ 3)\end{array}\right.$$

Modeling Electromagnetic Interactions

❖ Coupled-Circuit Phase Domain (CCPD) Model:

Electromagnetic Torque $T_e = \left(\frac{P}{2}\right) \cdot \frac{\partial W_c}{\partial \theta_r}$ P - number of magnetic poles

Energy Stored in Coupling Field

$$W_f = \frac{1}{2}\mathbf{i}_{abcs}^T\left(\mathbf{L}_s - L_{ls}\mathbf{I}\right)\mathbf{i}_{abcs} + \frac{1}{2}\mathbf{i}'^T_{qdr}\left(\mathbf{L}'_r - L'_{lr}\mathbf{I}\right)\mathbf{i}'_{qdr} + \mathbf{i}_{abcs}^T\left(\mathbf{L}'_{sr}\right)\mathbf{i}'_{qdr}$$

magnetically-linear system: $W_c = W_f$

$$T_e = \left(\frac{P}{2}\right)\left\{\frac{1}{2}\mathbf{i}_{abcs}^T\frac{\partial}{\partial\theta_r}\left[\mathbf{L}_s\left(\theta_r\right)\right]\mathbf{i}_{abcs} + \mathbf{i}_{abcs}^T\frac{\partial}{\partial\theta_r}\left[\mathbf{L}'_{sr}\left(\theta_r\right)\right]\mathbf{i}'_{qdr}\right\}$$

$$T_e = -\left(\frac{P}{2}\right)\left\{\frac{\left(L_{md} - L_{mq}\right)}{3}\left[\left(i_{as}^2 - \frac{1}{2}i_{bs}^2 - \frac{1}{2}i_{cs}^2 - i_{as}i_{bs} - i_{as}i_{cs} + 2i_{bs}i_{cs}\right)\sin\left(2\theta_r\right) + \right.\right.$$
$$\left. + \frac{\sqrt{3}}{2}\left(i_{bs}^2 + i_{cs}^2 - 2i_{as}i_{bs} + 2i_{as}i_{cs}\right)\cos\left(2\theta_r\right)\right] - L_{mq}\left(i'_{kq1} + i'_{kq2}\right)\left[\left(i_{as} - \frac{1}{2}i_{bs} - \frac{1}{2}i_{cs}\right)\sin\left(\theta_r\right) - \frac{\sqrt{3}}{2}\left(i_{bs} - i_{cs}\right)\cos\left(\theta_r\right)\right]$$
$$\left. + L_{md}\left(i'_{fd} + i'_{kd}\right)\left[\left(i_{as} - \frac{1}{2}i_{bs} - \frac{1}{2}i_{cs}\right)\cos\left(\theta_r\right) + \frac{\sqrt{3}}{2}\left(i_{bs} - i_{cs}\right)\cos\left(\theta_r\right)\right]\right\}$$

Modeling of Electric System

❖ *QD*0 Model:

➢ **Dynamic Equations:**

$$\frac{d}{dt}\begin{bmatrix} \boldsymbol{\lambda}_{abcs} \\ \boldsymbol{\lambda}'_{qdr} \end{bmatrix} = \begin{bmatrix} -\mathbf{R}_s & \\ & -\mathbf{R}'_r \end{bmatrix}\begin{bmatrix} \mathbf{i}_{abcs} \\ \mathbf{i}'_{qdr} \end{bmatrix} + \begin{bmatrix} \mathbf{v}_{abcs} \\ \mathbf{v}'_{qdr} \end{bmatrix} \qquad \begin{bmatrix} \boldsymbol{\lambda}_{abcs} \\ \boldsymbol{\lambda}'_{abcr} \end{bmatrix} = \begin{bmatrix} \mathbf{L}_s & \mathbf{L}'_{sr} \\ (2/3)\left(\mathbf{L}'_{sr}\right)^T & \mathbf{L}'_r \end{bmatrix}\begin{bmatrix} \mathbf{i}_{abcs} \\ \mathbf{i}'_{qdr} \end{bmatrix}$$

➢ Transform Stator variables to Rotor Reference Frame (RRF), using angle θ_r

$$\mathbf{K}_s(\theta_r) = \frac{2}{3}\begin{bmatrix} \cos(\theta_r) & \cos(\theta_r - 120°) & \cos(\theta_r + 120°) \\ \sin(\theta_r) & \sin(\theta_r - 120°) & \sin(\theta_r + 120°) \\ 1/2 & 1/2 & 1/2 \end{bmatrix}$$

$$\theta_r = \theta_r(0) + \int_0^t \omega_r dt \quad \Longrightarrow \quad \mathbf{f}_{qd0s} = \mathbf{K}_s(\theta_r)\mathbf{f}_{abcs}$$

✓ Rotor variables are already in the *qd* coordinates (i.e., rotor frame frame)

Modeling of Electric System

❖ *QD*0 Model:

- **Voltage Equations in phase *abc* coordinates:**

$$\begin{cases} \mathbf{v}_{abcs} = \mathbf{R}_s \mathbf{i}_{abcs} + \dfrac{d\boldsymbol{\lambda}_{abcs}}{dt} \\ \mathbf{v}'_{qdr} = \mathbf{R}'_r \mathbf{i}'_{qdr} + \dfrac{d\boldsymbol{\lambda}'_{qdr}}{dt} \end{cases}$$

- **Stator Voltage Equations in *qd*0 coordinates:**

$$\mathbf{v}_{qd0s} = \mathbf{R}_s \mathbf{i}_{qd0s} + \omega_r \boldsymbol{\lambda}_{dqs} + \frac{d\boldsymbol{\lambda}_{qd0s}}{dt}$$

where $\boldsymbol{\lambda}_{dqs} = \left[\lambda_{ds}, -\lambda_{qs}, 0\right]^T$

- **Expanded Form:**

$$\begin{cases} v_{qs} = r_s i_{qs} + \omega_r \lambda_{ds} + \dfrac{d\lambda_{qs}}{dt} \\ v_{ds} = r_s i_{ds} - \omega_r \lambda_{qs} + \dfrac{d\lambda_{ds}}{dt} \\ v_{0s} = r_s i_{0s} + \dfrac{d\lambda_{0s}}{dt} \end{cases}$$

$$\begin{cases} v'_{fd} = r'_{fd} i'_{fd} + \dfrac{d\lambda'_{fd}}{dt} \\ v'_{kd} = r'_{kd} i'_{kd} + \dfrac{d\lambda'_{kd}}{dt} \\ v'_{kq1} = r'_{kq1} i'_{kq1} + \dfrac{d\lambda'_{kq1}}{dt} \\ v'_{kq2} = r'_{kq2} i'_{kq2} + \dfrac{d\lambda'_{kq2}}{dt} \end{cases}$$

Modeling of Electric System

❖ ***QD*0 Model:**

- **Flux Linkage Equations in phase *abc* coordinates:**

$$\begin{bmatrix} \lambda_{abcs} \\ \lambda'_{qdr} \end{bmatrix} = \begin{bmatrix} \mathbf{L}_s & \mathbf{L}'_{sr} \\ (2/3)\left(\mathbf{L}'_{sr}\right)^T & \mathbf{L}'_r \end{bmatrix} \begin{bmatrix} \mathbf{i}_{abcs} \\ \mathbf{i}'_{qdr} \end{bmatrix}$$

$$\Rightarrow \begin{cases} \lambda_{abcs} = \mathbf{L}_s \mathbf{i}_{abcs} + \mathbf{L}'_{sr} \mathbf{i}'_{qdr} \\ \lambda'_{qdr} = \mathbf{L}'_r \mathbf{i}'_{qdr} + \left(2/3\right)\left(\mathbf{L}'_{sr}\right)^T \mathbf{i}_{abcs} \end{cases}$$

- **Flux Linkage Equations in *qd*0 coordinates:**

$$\boxed{\mathbf{f}_{qd0s} = \mathbf{K}_s\left(\theta_r\right)\mathbf{f}_{abcs}}$$

$$\Rightarrow \quad \lambda_{qd0s} =$$

Modeling of Electric System

❖ *QD*0 Model:

➢ **Flux Linkage Equations in *qd*0 coordinates:**

$$\boldsymbol{\lambda}_{qd0s} = \left[\mathbf{K}_s \mathbf{L}_s \mathbf{K}_s^{-1}\right]\mathbf{i}_{qd0s} + \left[\mathbf{K}_s \mathbf{L}'_{sr}\right]\mathbf{i}'_{qdr}$$

$$\boldsymbol{\lambda}'_{qdr} = \left[\mathbf{L}'_r\right]\mathbf{i}'_{qdr} + \left[\left(2/3\right)\left(\mathbf{L}'_{sr}\right)^T \mathbf{K}_s^{-1}\right]\mathbf{i}_{qd0s}$$

$$\mathbf{K}_s = \frac{2}{3}\begin{bmatrix} \cos(\theta_r) & \cos(\theta_r - 120°) & \cos(\theta_r + 120°) \\ \sin(\theta_r) & \sin(\theta_r - 120°) & \sin(\theta_r + 120°) \\ 1/2 & 1/2 & 1/2 \end{bmatrix}$$

$$\mathbf{L}_s = \left[\begin{array}{c:c:c} L_{ls} + L_A - L_B\cos 2(\theta_r) & -\frac{1}{2}L_A - L_B\cos 2\left(\theta_r - \frac{\pi}{3}\right) & -\frac{1}{2}L_A - L_B\cos 2\left(\theta_r + \frac{\pi}{3}\right) \\ -\frac{1}{2}L_A - L_B\cos 2\left(\theta_r - \frac{\pi}{3}\right) & L_{ls} + L_A - L_B\cos 2\left(\theta_r - \frac{2\pi}{3}\right) & -\frac{1}{2}L_A - L_B\cos 2(\theta_r + \pi) \\ -\frac{1}{2}L_A - L_B\cos 2\left(\theta_r + \frac{\pi}{3}\right) & -\frac{1}{2}L_A - L_B\cos 2(\theta_r + \pi) & L_{ls} + L_A - L_B\cos 2\left(\theta_r + \frac{2\pi}{3}\right) \end{array}\right]$$

$$\mathbf{L}'_{sr} = \left[\begin{array}{c:c:c:c} L_{mq}\cos(\theta_r) & L_{mq}\cos(\theta_r) & L_{md}\sin(\theta_r) & L_{md}\sin(\theta_r) \\ L_{mq}\cos\left(\theta_r - \frac{2\pi}{3}\right) & L_{mq}\cos\left(\theta_r - \frac{2\pi}{3}\right) & L_{md}\sin\left(\theta_r - \frac{2\pi}{3}\right) & L_{md}\sin\left(\theta_r - \frac{2\pi}{3}\right) \\ L_{mq}\cos\left(\theta_r + \frac{2\pi}{3}\right) & L_{mq}\cos\left(\theta_r + \frac{2\pi}{3}\right) & L_{md}\sin\left(\theta_r + \frac{2\pi}{3}\right) & L_{md}\sin\left(\theta_r + \frac{2\pi}{3}\right) \end{array}\right]$$

Modeling of Electric System

❖ *QD*0 Model:

➢ *qd*0 Inductance Matrices:

$$\boldsymbol{\lambda}_{qd0s} = \left[\mathbf{K}_s\mathbf{L}_s\mathbf{K}_s^{-1}\right]\mathbf{i}_{qd0s} + \left[\mathbf{K}_s\mathbf{L}'_{sr}\right]\mathbf{i}'_{qdr}$$
$$\boldsymbol{\lambda}'_{qdr} = \left[\mathbf{L}'_r\right]\mathbf{i}'_{qdr} + \left[\left(2/3\right)\left(\mathbf{L}'_{sr}\right)^T\mathbf{K}_s^{-1}\right]\mathbf{i}_{qd0s}$$

$$\mathbf{K}_s\mathbf{L}_s\mathbf{K}_s^{-1} = \left|\begin{array}{c:c:c} L_{ls}+L_{mq} & & \\ & L_{ls}+L_{md} & \\ & & L_{ls} \end{array}\right| \qquad \mathbf{K}_s\mathbf{L}'_{sr} = \left|\begin{array}{c:c:c:c} L_{mq} & L_{mq} & & \\ & & L_{md} & L_{md} \\ & & & \end{array}\right|$$

$$\mathbf{L}'_r = \left[\begin{array}{c:c:c:c} L'_{lkq1}+L_{mq} & L_{mq} & & \\ L_{mq} & L'_{kq2}+L_{mq} & & \\ & & L'_{lfd}+L_{md} & L_{md} \\ & & L_{md} & L_{lkd}+L_{md} \end{array}\right] \qquad \left(2/3\right)\left(\mathbf{L}'_{sr}\right)^T\mathbf{K}_s^{-1} = \left[\begin{array}{c:c:c} L_{mq} & & \\ L_{mq} & & \\ & L_{md} & \\ & L_{md} & \end{array}\right]$$

➢ Expanded Form:

$$\begin{cases} \lambda_{qs} = L_{ls}i_{qs} + L_{mq}\left(i_{qs} + i'_{kq1} + i'_{kq2}\right) \\ \lambda_{ds} = L_{ls}i_{ds} + L_{md}\left(i_{ds} + i'_{fd} + i'_{kd}\right) \\ \lambda_{0s} = L_{ls}i_{0s} \end{cases}$$

$$\begin{cases} \lambda'_{kq1} = L'_{lkq1}i'_{kq1} + L_{mq}\left(i_{qs} + i'_{kq1} + i'_{kq2}\right) \\ \lambda'_{kq2} = L'_{lkq2}i'_{kq2} + L_{mq}\left(i_{qs} + i'_{kq1} + i'_{kq2}\right) \\ \lambda'_{fd} = L'_{lfd}i'_{fd} + L_{md}\left(i_{ds} + i'_{fd} + i'_{kd}\right) \\ \lambda'_{kd} = L'_{lkd}i'_{kd} + L_{md}\left(i_{ds} + i'_{fd} + i'_{kd}\right) \end{cases}$$

Modeling of Electric System

❖ QD0 Model:

➢ Winding Voltage Equations:

$$v_{qs} = r_s i_{qs} + \omega_r \lambda_{ds} + \frac{d\lambda_{qs}}{dt}$$

$$v_{ds} = r_s i_{ds} - \omega_r \lambda_{qs} + \frac{d\lambda_{ds}}{dt}$$

$$v_{0s} = r_s i_{0s} + \frac{d\lambda_{0s}}{dt}$$

$$v'_{kq1} = r'_{kq1} i'_{kq1} + \frac{d\lambda'_{kq1}}{dt}$$

$$v'_{kq2} = r'_{kq2} i'_{kq2} + \frac{d\lambda'_{kq2}}{dt}$$

$$v'_{fd} = r'_{fd} i'_{fd} + \frac{d\lambda'_{fd}}{dt}$$

$$v'_{kd} = r'_{kd} i'_{kd} + \frac{d\lambda'_{kd}}{dt}$$

➢ Flux Linkage Equations:

$$\lambda_{qs} = L_{ls} i_{qs} + L_{mq}\left(i_{qs} + i'_{kq1} + i'_{kq2}\right)$$

$$\lambda_{ds} = L_{ls} i_{ds} + L_{md}\left(i_{ds} + i'_{fd} + i'_{kd}\right)$$

$$\lambda_{0s} = L_{ls} i_{0s}$$

$$\lambda'_{kq1} = L'_{lkq1} i'_{kq1} + L_{mq}\left(i_{qs} + i'_{kq1} + i'_{kq2}\right)$$

$$\lambda'_{kq2} = L'_{lkq2} i'_{kq2} + L_{mq}\left(i_{qs} + i'_{kq1} + i'_{kq2}\right)$$

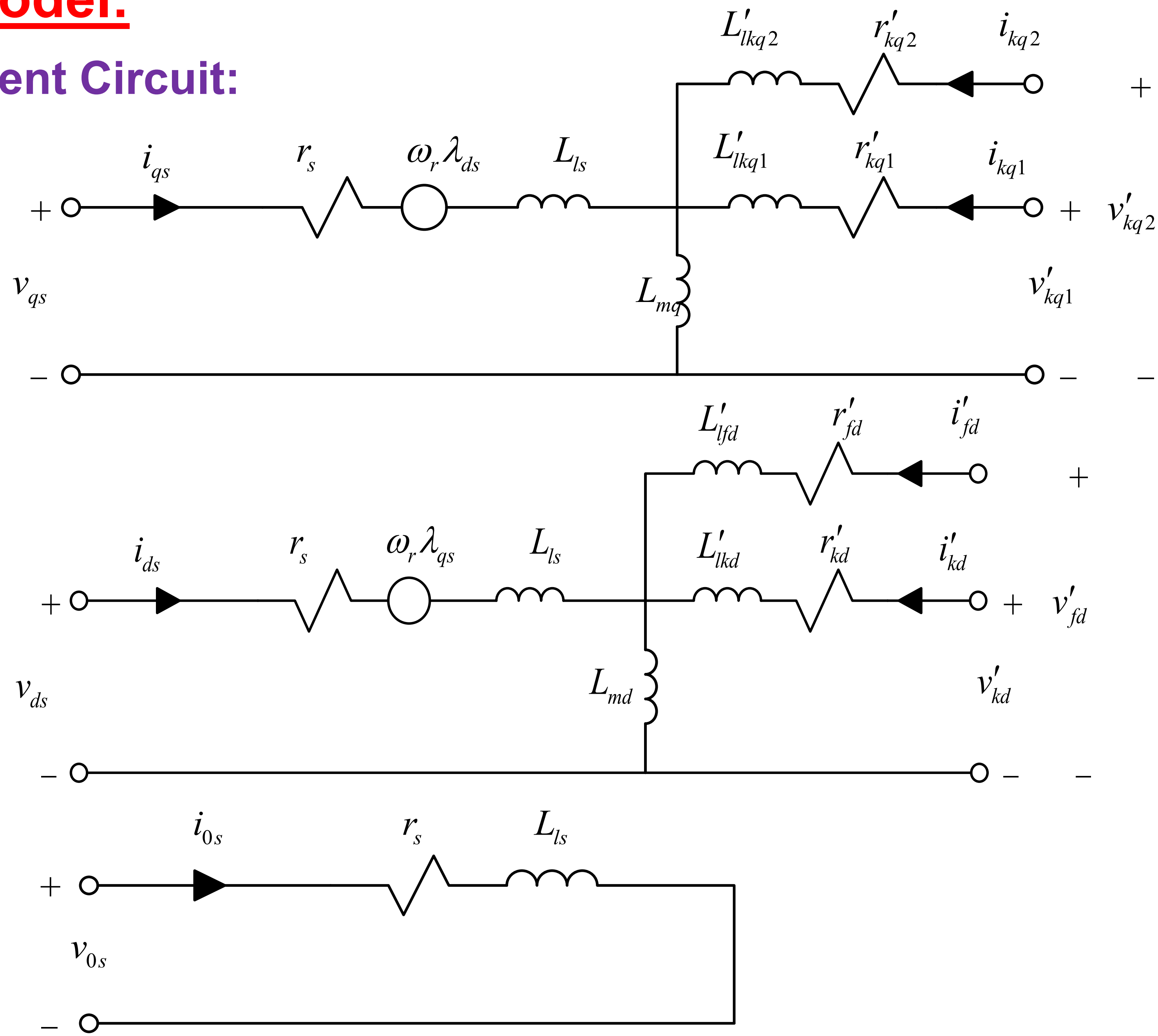
$$\lambda'_{fd} = L'_{lfd} i'_{fd} + L_{md}\left(i_{ds} + i'_{fd} + i'_{kd}\right)$$

$$\lambda'_{kd} = L'_{lkd} i'_{kd} + L_{md}\left(i_{ds} + i'_{fd} + i'_{kd}\right)$$

Modeling of Electric System

❖ *QD*0 Model:

➢ **Equivalent Circuit:**



Modeling Electromagnetic Interactions

❖ *QD*0 Model:

Electromagnetic Torque:

$$T_e = \left(\frac{P}{2}\right)\left\{\frac{1}{2}\mathbf{i}_{abcs}^T \frac{\partial}{\partial\theta_r}\left[\mathbf{L}_s(\theta_r)\right]\mathbf{i}_{abcs} + \mathbf{i}_{abcs}^T \frac{\partial}{\partial\theta_r}\left[\mathbf{L}'_{sr}(\theta_r)\right]\mathbf{i}'_{qdr}\right\}$$

$$\mathbf{i}_{abcs} = \mathbf{K}_s^{-1}\mathbf{i}_{qd0s}$$

$$\mathbf{L}_s = \begin{bmatrix} L_{ls} + L_A - L_B\cos 2(\theta_r) & -\frac{1}{2}L_A - L_B\cos 2\left(\theta_r - \frac{\pi}{3}\right) & -\frac{1}{2}L_A - L_B\cos 2\left(\theta_r + \frac{\pi}{3}\right) \\ -\frac{1}{2}L_A - L_B\cos 2\left(\theta_r - \frac{\pi}{3}\right) & L_{ls} + L_A - L_B\cos 2\left(\theta_r - \frac{2\pi}{3}\right) & -\frac{1}{2}L_A - L_B\cos 2(\theta_r + \pi) \\ -\frac{1}{2}L_A - L_B\cos 2\left(\theta_r + \frac{\pi}{3}\right) & -\frac{1}{2}L_A - L_B\cos 2(\theta_r + \pi) & L_{ls} + L_A - L_B\cos 2\left(\theta_r + \frac{2\pi}{3}\right) \end{bmatrix}$$

$$\mathbf{K}_s = \frac{2}{3}\begin{bmatrix} \cos(\theta_r) & \cos(\theta_r - 120^\circ) & \cos(\theta_r + 120^\circ) \\ \sin(\theta_r) & \sin(\theta_r - 120^\circ) & \sin(\theta_r + 120^\circ) \\ 1/2 & 1/2 & 1/2 \end{bmatrix}$$

$$\mathbf{L}'_{sr} = \begin{bmatrix} L_{mq}\cos\theta_r & L_{mq}\cos\theta_r & L_{md}\sin\theta_r & L_{md}\sin\theta_r \\ L_{mq}\cos\left(\theta_r - \frac{2\pi}{3}\right) & L_{mq}\cos\left(\theta_r - \frac{2\pi}{3}\right) & L_{md}\sin\left(\theta_r - \frac{2\pi}{3}\right) & L_{md}\sin\left(\theta_r - \frac{2\pi}{3}\right) \\ L_{mq}\cos\left(\theta_r + \frac{2\pi}{3}\right) & L_{mq}\cos\left(\theta_r + \frac{2\pi}{3}\right) & L_{md}\sin\left(\theta_r + \frac{2\pi}{3}\right) & L_{md}\sin\left(\theta_r + \frac{2\pi}{3}\right) \end{bmatrix}$$

$$\begin{aligned} T_e &= \left(\frac{P}{2}\right)\left(\frac{3}{2}\right)\left[L_{md}\left(i_{ds} + i'_{fd} + i'_{kd}\right)i_{qs} - L_{mq}\left(i_{qs} + i'_{kq1} + i'_{kq2}\right)i_{ds}\right] \\ &= \left(\frac{P}{2}\right)\left(\frac{3}{2}\right)\left(\lambda_{ds}i_{qs} - \lambda_{qs}i_{ds}\right) = \left(\frac{P}{2}\right)\left(\frac{3}{2}\right)\frac{1}{\omega_b}\left(\psi_{ds}i_{qs} - \psi_{qs}i_{ds}\right) \end{aligned}$$

Modeling of Electric System

❖ *QD*0 Model:

➢ Winding Voltage Equations:

$$v_{qs} = r_s i_{qs} + \omega_r \lambda_{ds} + \frac{d\lambda_{qs}}{dt}$$

$$v_{ds} = r_s i_{ds} - \omega_r \lambda_{qs} + \frac{d\lambda_{ds}}{dt}$$

$$v_{0s} = r_s i_{0s} + \frac{d\lambda_{0s}}{dt}$$

$$v'_{kq1} = r'_{kq1} i'_{kq1} + \frac{d\lambda'_{kq1}}{dt}$$

$$v'_{kq2} = r'_{kq2} i'_{kq2} + \frac{d\lambda'_{kq2}}{dt}$$

$$v'_{fd} = r'_{fd} i'_{fd} + \frac{d\lambda'_{fd}}{dt}$$

$$v'_{kd} = r'_{kd} i'_{kd} + \frac{d\lambda'_{kd}}{dt}$$

➢ Flux Linkage Equations:

$$\lambda_{qs} = L_{ls} i_{qs} + L_{mq}\left(i_{qs} + i'_{kq1} + i'_{kq2}\right)$$

$$\lambda_{ds} = L_{ls} i_{ds} + L_{md}\left(i_{ds} + i'_{fd} + i'_{kd}\right)$$

$$\lambda_{0s} = L_{ls} i_{0s}$$

$$\lambda'_{kq1} = L'_{lkq1} i'_{kq1} + L_{mq}\left(i_{qs} + i'_{kq1} + i'_{kq2}\right)$$

$$\lambda'_{kq2} = L'_{lkq2} i'_{kq2} + L_{mq}\left(i_{qs} + i'_{kq1} + i'_{kq2}\right)$$

$$\lambda'_{fd} = L'_{lfd} i'_{fd} + L_{md}\left(i_{ds} + i'_{fd} + i'_{kd}\right)$$

$$\lambda'_{kd} = L'_{lkd} i'_{kd} + L_{md}\left(i_{ds} + i'_{fd} + i'_{kd}\right)$$

➢ Let's use Flux Linkages per-second and Reactances

$$\psi = \lambda \times \omega_b \qquad x = \omega_b \times L$$

Modeling of Electric System

❖ ***QD*0 Model:**

➢ State Equations:

$$\frac{d\psi_{qs}}{dt} = \omega_b\left(-r_s i_{qs} - \frac{\omega_r}{\omega_b}\psi_{ds} + v_{qs}\right)$$

$$\frac{d\psi_{ds}}{dt} = \omega_b\left(-r_s i_{ds} + \frac{\omega_r}{\omega_b}\psi_{qs} + v_{ds}\right)$$

$$\frac{d\psi_{0s}}{dt} = \omega_b\left(-r_s i_{0s} + v_{0s}\right)$$

$$\frac{d\psi'_{kq1}}{dt} = \omega_b\left(-r'_{kq1} i'_{kq1}\right)$$

$$\frac{d\psi'_{kq2}}{dt} = \omega_b\left(-r'_{kq2} i'_{kq2}\right)$$

$$\frac{d\psi'_{fd}}{dt} = \omega_b\left(-r'_{fd} i'_{fd} + v_{fd}\right)$$

$$\frac{d\psi'_{kd}}{dt} = \omega_b\left(-r'_{kd} i'_{kd}\right)$$

➢ Flux Equations:

$$\psi_{qs} = x_{ls} i_{qs} + \psi_{mq}$$

$$\psi_{ds} = x_{ls} i_{ds} + \psi_{md}$$

$$\psi_{0s} = x_{ls} i_{0s}$$

$$\psi'_{kq1} = x'_{lkq1} i'_{kq1} + \psi_{mq}$$

$$\psi'_{kq2} = x'_{lkq2} i'_{kq2} + \psi_{mq}$$

$$\psi'_{fd} = x'_{lfd} i'_{fd} + \psi_{md}$$

$$\psi'_{kd} = x'_{lkd} i'_{kd} + \psi_{md}$$

$$\psi_{mq} = x_{mq}\left(i_{qs} + i'_{kq1} + i'_{kq2}\right)$$

$$\psi_{md} = x_{md}\left(i_{ds} + i'_{fd} + i'_{kd}\right)$$

magnetizing Flux Linkages per-second

Modeling of Electric System

❖ *QD*0 Model:

➢ Output Equations:

Stator currents:

$$i_{qs} = \frac{\psi_{qs}}{x_{ls}} - \frac{\psi_{mq}}{x_{ls}} \qquad i_{ds} = \frac{\psi_{ds}}{x_{ls}} - \frac{\psi_{md}}{x_{ls}} \qquad i_{0s} = \frac{\psi_{0s}}{x_{ls}}$$

Rotor currents:

$$i'_{kq1} = \frac{\left(\psi'_{kq1} - \psi_{mq}\right)}{x'_{lkq1}} \qquad i'_{kq2} = \frac{\left(\psi'_{kq2} - \psi_{mq}\right)}{x'_{lkq2}} \qquad i'_{fd} = \frac{\left(\psi'_{fd} - \psi_{md}\right)}{x'_{lfd}} \qquad i'_{kd} = \frac{\left(\psi'_{kd} - \psi_{md}\right)}{x'_{lkd}}$$

$$\psi_{mq} = x_{mq}\left(i_{qs} + i'_{kq1} + i'_{kq2}\right)$$

$$\psi_{md} = x_{md}\left(i_{ds} + i'_{fd} + i'_{kd}\right)$$

$$\psi_{mq} = x_{mq}\left(\frac{\psi_{qs}}{x_{ls}} - \frac{\psi_{mq}}{x_{ls}} + \frac{\psi'_{kq1}}{x'_{lkq1}} - \frac{\psi_{mq}}{x'_{lkq1}} + \frac{\psi'_{kq2}}{x'_{lkq2}} - \frac{\psi_{mq}}{x'_{lkq2}}\right)$$

$$\psi_{mq} = x_{aq}\left(\frac{\psi_{qs}}{x_{ls}} + \frac{\psi'_{kq1}}{x'_{lkq1}} + \frac{\psi'_{kq2}}{x'_{lkq2}}\right) \qquad x_{aq} = \left(\frac{1}{x_{ls}} + \frac{1}{x'_{lkq1}} + \frac{1}{x'_{lkq2}} + \frac{1}{x_{mq}}\right)^{-1}$$

Similarly, for *d*-axis

$$\psi_{md} = x_{ad}\left(\frac{\psi_{ds}}{x_{ls}} + \frac{\psi'_{fd}}{x'_{lfd}} + \frac{\psi'_{kd}}{x'_{lkd}}\right) \qquad x_{ad} = \left(\frac{1}{x_{ls}} + \frac{1}{x'_{lfd}} + \frac{1}{x'_{lkd}} + \frac{1}{x_{md}}\right)^{-1}$$

Modeling of Electric System

❖ Per-Unit System:

$$f_{pu} = \frac{f_{actual}}{f_{base}} \qquad f_{actual} = f_{pu} \cdot f_{base}$$

$$f \in \{i, v, \lambda, \psi, P, T\}$$

f_{base} is the assumed base value

for *abc* variables	for *qd* variables
Base Voltages:	
$v_{base} = v_{B(abc)} = v_{rms}$	$v_{base} = v_{B(qd)} = v_{peak}$
rms of phase voltage	peak value → $\boxed{v_{B(qd)} = \sqrt{2} \times v_{B(abc)}}$
Base Currents:	
$i_{base} = i_{B(abc)} = i_{rms}$	$i_{base} = i_{B(qd)} = i_{peak}$
rms of phase current	peak value → $\boxed{i_{B(qd)} = \sqrt{2} \times i_{B(abc)}}$
Base Power:	→ $\boxed{P_{B(abc)} = P_{B(qd)} = P_B}$
$P_{B(abc)} = (3)\, v_{B(abc)} \times i_{B(abc)}$	$P_{B(qd)} = (3/2)\, v_{B(qd)} \times i_{B(qd)}$
Base Impedance:	→ $\boxed{z_{B(abc)} = z_{B(qd)} = z_B}$
$z_{B(abc)} = \frac{v_{B(abc)}}{I_{B(abc)}} = 3 \times \frac{v_{B(abc)}^2}{P_B}$	$z_{B(qd)} = \frac{v_{B(qd)}}{I_{B(qd)}} = \left(\frac{3}{2}\right) \frac{v_{B(qd)}^2}{P_B}$

Modeling of Electric System

❖ Per-Unit System:

➢ **Base Torque:**

$$T_B = \left(\frac{P}{2}\right)\frac{P_B}{\omega_B} \Rightarrow T_B = \left(\frac{P}{2}\right)\left(\frac{1}{\omega_b}\right)\left(\frac{3}{2}\right)v_{B(qd)} \times i_{B(qd)}$$

Recall:

$$T_e = \left(\frac{P}{2}\right)\left(\frac{3}{2}\right)\left(\frac{1}{\omega_b}\right)\left(\psi_{ds}i_{qs} - \psi_{qs}i_{ds}\right) \Rightarrow T_e^{pu} =$$

$$\boxed{T_e^{pu} = \frac{T_e}{T_B} = \psi_{ds}^{pu}i_{qs}^{pu} - \psi_{qs}^{pu}i_{ds}^{pu}}$$

➢ **Mechanical Dynamics:**

$$J\frac{d\omega_{rm}}{dt} = J\left(\frac{2}{P}\right)\frac{d\omega_r}{dt} = T_e - T_m \Rightarrow J\left(\frac{2}{P}\right)^2\frac{\omega_b}{P_B}\frac{d\omega_r}{dt} = T_e^{pu} - T_m^{pu}$$

$$\Rightarrow \left\{J\left(\frac{2}{P}\right)^2\frac{\omega_b^{\;2}}{P_B}\right\}\frac{d\omega_r^{pu}}{dt} = T_e^{pu} - T_m^{pu} \Rightarrow \boxed{2H\frac{d\omega_r^{pu}}{dt} = T_e^{pu} - T_m^{pu}}$$

Define **Inertia Constant**

$$\boxed{H = \frac{1}{2}J\left(\frac{2}{P}\right)^2\frac{\omega_b^2}{P_B} \quad [s]}$$